A hypergraph Turán problem with no stability

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Abstract

A fundamental barrier in extremal hypergraph theory is the presence of many near-extremal constructions with very different structure. Indeed, the notorious Turán problem for the complete triple system on four points most likely exhibits this phenomenon. We construct a finite family of triple systems $\mathcal{M}$, determine its Turán number, and prove that there are two near-extremal $\mathcal{M}$-free constructions that are far from each other in edit-distance. This is the first extremal result for a hypergraph family that fails to have a corresponding stability theorem.

1 Introduction

Let $r \geq 3$ and $\mathcal{F}$ be a family of r-uniform graphs (henceforth r-graphs). An r-graph $\mathcal{H}$ is $\mathcal{F}$-free if it contains no member of $\mathcal{F}$ as a subgraph. The Turán number $\mathrm{ex}(n, \mathcal{F})$ of $\mathcal{F}$ is the maximum number of edges in an $\mathcal{F}$-free r-graph on n vertices. The Turán density $\pi(\mathcal{F})$ of $\mathcal{F}$ is defined as $\pi(\mathcal{F}) := \lim_{n\to\infty} \mathrm{ex}(n, \mathcal{F})/\binom{n}{r}$. The study of $\mathrm{ex}(n, \mathcal{F})$ is perhaps the central topic in extremal graph and hypergraph theory.

Much is known about $\mathrm{ex}(n, \mathcal{F})$ when $r = 2$ and one of the most famous results in this regard is Turán's theorem, which states that for $\ell \geq 2$ the Turán number $\mathrm{ex}(n, K_{\ell+1})$ is uniquely achieved by $T(n, \ell)$ which is the ℓ-partite graph on n vertices with the maximum number of edges.

For $\ell > r \geq 3$, let K_ℓ^r be the complete r-graph on ℓ vertices. Extending Turán's theorem to hypergraphs (i.e. $r \geq 3$) is a major problem. Indeed, the problem of determining $\pi(K_\ell^r)$ was raised by Turán [24] and is still wide open. Erdős offered \$500 for the determination of any $\pi(K_\ell^r)$ with $\ell > r \geq 3$ and \$1000 for the determination of all $\pi(K_\ell^r)$ with $\ell > r \geq 3$.

Conjecture 1.1 (Turán, see [24])**.** $\pi(K_\ell^3) = 1 - \left(\frac{2}{\ell-1}\right)^2$.

The case $\ell = 4$ above, which states that $\pi(K_4^3) = 5/9$ has generated a lot of interest and activity over the years. Many constructions (e.g. see [1, 11, 4]) are known to achieve the value in Conjecture 1.1 for $\ell = 4$, and that is perhaps one of the reasons why it is so challenging. On the other hand, successively better upper bounds for $\pi(K_4^3)$ were obtained by de Caen [3], Giraud (see [2]), Chung and Lu [2], and Razborov [21]. The current record is $\pi(K_4^3) \leq 0.561666$, which was obtained by Razborov [21] using the Flag Algebra machinery.

Many families $\mathcal{F}$ have the property that there is a unique $\mathcal{F}$-free graph (or hypergraph) $\mathcal{G}$ on n vertices achieving $\mathrm{ex}(n, \mathcal{F})$, and moreover, any $\mathcal{F}$-free graph (or hypergraph) $\mathcal{H}$ of

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size close to $\mathrm{ex}(n, \mathcal{F})$ can be transformed to $\mathcal{G}$ by deleting and adding very few edges. Such a property is called the stability of $\mathcal{F}$. The first stability theorem was proved independently by Erdős and Simonovits [23].

Theorem 1.2 (Erdős, Simonovits, see [23]). *Let $\ell \geq 2$. Then for every $\delta > 0$ there exists an $\epsilon > 0$ and n_0 such that the following statement holds for all $n \geq n_0$. Every $K_{\ell+1}$-free graph on n vertices with at least $(1-\epsilon)t(n,\ell)$ edges can be transformed to $T(n,\ell)$ by deleting and adding at most δn^2 edges.*

The stability phenomenon has been used to determine $\mathrm{ex}(n, \mathcal{F})$ exactly in many cases. It was first used by Simonovits in [23] to determine $\mathrm{ex}(n, \mathcal{F})$ exactly for all color-critical graphs and large n, and then by several authors (e.g. see [7, 9, 10, 17, 20]) to prove exact results for hypergraphs.

However, there are many Turán problems for hypergraphs that (perhaps) do not have the stability property. The example K_4^3 we mentioned before was shown to have exponentially many extremal constructions [11]. These can be used to show that K_4^3 does not have the stability property (assuming Conjecture 1.1 is true). For K_ℓ^3 with $\ell \geq 5$, different near-extremal constructions were given by Sidorenko [22], and Keevash and the second author [8]. So K_ℓ^3 does not have stability as well (assuming Conjecture 1.1 is true).

Another example in which the forbidden family consists of infinitely many 3-graphs arises from the following longstanding conjecture due to Erdős and Sós. For an r-graph $\mathcal{H}$ we use $V(\mathcal{H})$ to denote its vertex set and $v(\mathcal{H}) = |V(\mathcal{H})|$. The link of $v \in V(\mathcal{H})$ is

$$L_{\mathcal{H}}(v) := \left\{ A \in \binom{V(\mathcal{H})}{r-1} : \{v\} \cup A \in \mathcal{H} \right\}.$$

We will omit the subscript $\mathcal{H}$ if it is clear from context.

Conjecture 1.3 (Erdős and Sós, see [5]). *Let $\mathcal{H}$ be a 3-graph with n-vertices. If $L(v)$ is bipartite for all $v \in V(\mathcal{H})$, then $|\mathcal{H}| \leq (1/4 + o(1))\binom{n}{3}$.*

If Conjecture 1.3 is true, then it also does not have the stability property as there are several different near-extremal constructions given in [18].

The absence of stability seems to be a fundamental barrier in determining the Turán numbers of some families. Indeed, the Turán numbers of the examples we presented above are not known, even asymptotically, and in fact, no Turán number of a family without the stability property has been determined.

This paper provides the first such example. We construct a family $\mathcal{M}$ of 3-graphs, prove that $\mathcal{M}$ does not have the stability property, and determine $\pi(\mathcal{M})$, and even $\mathrm{ex}(n, \mathcal{F})$ for infinitely many n (Theorems 1.7 and 1.12). In order to state our results formally, we need some definitions.

Definition 1.4. *Let $r \geq 2$ and $\mathcal{H}$ be an r-graph. The transversal number of $\mathcal{H}$ is*

$$\tau(\mathcal{H}) := \min\left\{|S| : S \subset V(\mathcal{H}) \text{ such that } S \cap E \neq \emptyset \text{ for all } E \in \mathcal{H}\right\}.$$

We set $\tau(\mathcal{H}) = 0$ if $\mathcal{H}$ is an empty graph.

Let $\ell \geq r \geq 2$ and $\mathcal{K}_{\ell+1}^r$ be the collection of all r-graphs F with at most $\binom{\ell+1}{2}$ edges such that for some $(\ell+1)$-set S, which will be called the core of F, every pair $\{u, v\} \subset S$ is covered by an edge in F. Let $V_1 \cup \cdots \cup V_\ell$ be a partition of $[n] := \{1, \ldots, n\}$ with each V_i of size either $\lfloor n/\ell \rfloor$ or $\lceil n/\ell \rceil$. The generalized Turán graph $T_r(n,\ell)$ is the collection of all r-subsets of $[n]$ that have at most one vertex in each V_i. Let $t_r(n,\ell) = |T_r(n,\ell)|$. It was shown by the second author [15] that $\mathrm{ex}(n, \mathcal{K}_{\ell+1}^r) = t_r(n,\ell)$ and $T_r(n,\ell)$ is the unique $\mathcal{K}_{\ell+1}^r$-free r-graph on n vertices with exactly $t_r(n,\ell)$ edges.

Suppose that $\mathcal{T}$ is an r-graph on s vertices and $t = (t_1, \ldots, t_s)$ with each t_i a positive integer. Then the blow up $\mathcal{T}(t)$ of $\mathcal{T}$ is obtained from $\mathcal{T}$ by replacing each vertex i by a set of size t_i, and replacing every edge in $\mathcal{T}$ by the corresponding complete r-partite r-graph.

Definition 1.5. *Let $|A| = \lfloor n/3 \rfloor$ and $|B| = \lceil 2n/3 \rceil$ with $A \cap B = \emptyset$. Define*

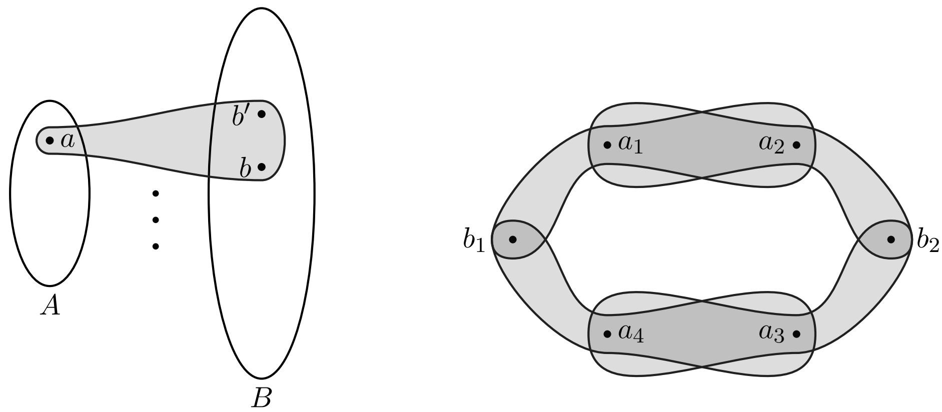
$$\mathcal{G}_n^1 := \left\{ abb' : a \in A \text{ and } \{b, b'\} \subset B \right\}.$$

Let $\mathcal{G}_6^2$ be the 3-graph on six vertices $\{a_1, a_2, a_3, a_4, b_1, b_2\}$ whose complement is

$$\overline{\mathcal{G}_6^2} := \{a_1a_2b_1, a_1a_2b_2, a_3a_4b_1, a_3a_4b_2\}.$$

For $n > 6$ let $\mathcal{G}_n^2$ be a 3-graph on n vertices which is a blow up of $\mathcal{G}_6^2$ with the maximum number of edges.

Simple calculations show that each part in $\mathcal{G}_n^2$ has size either $\lfloor n/6 \rfloor$ or $\lceil n/6 \rceil$. For $i = 1, 2$, let $g_i(n) = |\mathcal{G}_n^i|$ and note that $\lim_{n\to\infty} g_i(n)/n^3 = 2/27$.



(a) The 3-graph $\mathcal{G}_n^1$. (b) The complement of $\mathcal{G}_6^2$.

Figure 1: $\mathcal{G}^1$ and $\overline{\mathcal{G}_6^2}$.

Definition 1.6. *The family $\mathcal{M}$ is the union of the following three families.*

(a) M_1 is the set containing the complete 3-graph on five vertices with one edge removed, i.e. $M_1 = \left\{K_5^{3-}\right\}$.

(b) M_2 is the collection of all 3-graphs in $\mathcal{K}_7^3$ whose induced subgraph on the core has transversal number at least two.

(c) M_3 is the collection of all 3-graphs F on six vertices such that $F \not\subset \mathcal{G}_n^1$ and $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$.

Our first result is about the Turán number of $\mathcal{M}$.

Theorem 1.7. *The inequality $ex(n, \mathcal{M}) \leq 2n^3/27$ holds for all $n \geq 1$. Moreover, equality holds whenever n is a multiple of six.*

For an r-graph $\mathcal{H}$ the shadow of $\mathcal{H}$ is

$$\partial\mathcal{H} := \left\{ A \in \binom{V(\mathcal{H})}{r-1} : \exists B \in \mathcal{H} \text{ such that } A \subset B \right\}.$$

Note that both $\mathcal{G}_n^1$ and $\mathcal{G}_n^2$ are $\mathcal{M}$-free and $g_1(n) \sim g_2(n) \sim 2n^3/27$. Moreover, it is easy to see that transforming $\mathcal{G}_n^1$ to $\mathcal{G}_n^2$ requires us to delete and add $\Omega(n^3)$ edges. Indeed, in order to transform $\partial\mathcal{G}_n^1$ to $\partial\mathcal{G}_n^2$ we need to remove $\Omega(n^2)$ edges from $\partial\mathcal{G}_n^1$. Since every edge in $\partial\mathcal{G}_n^1$ is covered by $\Omega(n)$ edges in $\mathcal{G}_n^1$, we need to remove at least $\Omega(n^3)$ edges from $\mathcal{G}_n^1$ before getting $\mathcal{G}_n^2$. So this proves that $\mathcal{M}$ does not have the stability property (in the sense of Theorem 1.2).

In order to capture the structural property of families with more than one near-extremal structure, the second author [16] introduced the concept of t-stable families, which is an extension of the classical definition of stability.

Definition 1.8 (t-stable, see [16]). *Let $r \geq 2$ and $t \geq 1$ and $\mathcal{F}$ be a family of r-graphs. Then $\mathcal{F}$ is t-stable if there exists m_0 and r-graphs $\mathcal{H}_m^1, \ldots, \mathcal{H}_m^t$ on m vertices for every $m \geq m_0$ such that the following holds. For every $\delta > 0$, there exists $\epsilon > 0$ and n_0 such that for all $n \geq n_0$, if $\mathcal{H}$ is an $\mathcal{F}$-free r-graph on n vertices with*

$$|\mathcal{H}| > (1-\epsilon)\mathrm{ex}(n, \mathcal{F}),$$

then $\mathcal{H}$ can be transformed to some $\mathcal{H}_n^i$ by adding and removing at most $\delta|\mathcal{H}|$ edges. Say $\mathcal{F}$ is stable if it is 1-stable.

According to the definition, if a family $\mathcal{F}$ is t-stable, then it is s-stable for all $s > t$, since we can get s constructions $\mathcal{G}_m^1, \ldots, \mathcal{G}_m^s$ by simply duplicating some $\mathcal{G}_m^i \in \{\mathcal{G}_m^1, \ldots, \mathcal{G}_m^t\}$ $s-t$ times. However, we are actually interested in the minimum integer t such that $\mathcal{F}$ is t-stable. Therefore, we introduce the stability number of a family $\mathcal{F}$.

Definition 1.9 (Stability number). *Let $\mathcal{F}$ be a family of r-graphs. The stability number of $\mathcal{F}$, denoted by $\xi(\mathcal{F})$, is the minimum integer t such that $\mathcal{F}$ is t-stable. If there is no such integer t, then we let $\xi(\mathcal{F}) = \infty$.*

Note that in [19] Pikhurko also gave a definition for t-stable families and it is essentially the same as Definition 1.8. Roughly speaking, a family $\mathcal{F}$ is t-stable if there exist t near-extremal constructions, and every $\mathcal{F}$-free graph (or hypergraph) of size close to $\mathrm{ex}(n, \mathcal{F})$ is structurally close to one of these near-extremal constructions. Although the concept of t-stable families was raised over a decade ago, no example of t-stable families are known for any $t \geq 2$ before this work.

Our next result gives further detail about near-extremal $\mathcal{M}$-free constructions by showing that $\mathcal{M}$ is 2-stable with respect to $\mathcal{G}_n^1$ and $\mathcal{G}_n^2$. More precisely, it shows that $\xi(\mathcal{M}) = 2$.

Definition 1.10. *Let $\mathcal{H}$ be a 3-graph. Then $\mathcal{H}$ is called semibipartite if $V(\mathcal{H})$ has a partition $A \cup B$ such that $|E \cap A| = 1$ and $|E \cap B| = 2$ for all $E \in \mathcal{H}$, and $\mathcal{H}$ is called $\mathcal{G}_6^2$-colorable if it is a subgraph of a blow up of $\mathcal{G}_6^2$.*

With some calculations one can get the following observation.

Observation 1.11. *Let $\mathcal{H}$ be a 3-graph on n-vertices. If $\mathcal{H}$ is semibipartite, then $|\mathcal{H}| \leq g_1(n)$. If $\mathcal{H}$ is $\mathcal{G}_6^2$-colorable, then $|\mathcal{H}| \leq g_2(n)$.*

Theorem 1.12 (2-stability). *For every $\delta > 0$ there exists $\epsilon > 0$ and n_0 such that the following holds for all $n \geq n_0$. Every $\mathcal{M}$-free 3-graph on n vertices with at least $2n^3/27 - \epsilon n^3$ edges can be transformed to a 3-graph that is either semibipartite or $\mathcal{G}_6^2$-colorable by removing at most δn vertices. In other words, $\xi(\mathcal{M}) = 2$.*

Note that Theorem 1.12 is stronger than the requirement in the definition of 2-stability since removing at most δn vertices implies that the number of edges removed is at most δn^3 but not vice versa.

Let $\mathcal{H}$ be an r-graph on n vertices. The edge density of $\mathcal{H}$ is $d(\mathcal{H}) := |\mathcal{H}|/\binom{n}{r}$ and the shadow density of $\mathcal{H}$ is $d(\partial\mathcal{H}) := |\partial\mathcal{H}|/\binom{n}{r-1}$. The feasible region $\Omega(\mathcal{F})$ of $\mathcal{F}$ is the set of points $(x,y) \in [0,1]^2$ such that there exists a sequence of $\mathcal{F}$-free r-graphs $(\mathcal{H}_k)_{k=1}^{\infty}$ with $\lim_{k\to\infty} v(\mathcal{H}_k) = \infty$, $\lim_{k\to\infty} d(\partial\mathcal{H}_k) = x$ and $\lim_{k\to\infty} d(\mathcal{H}_k) = y$. We introduced this notation recently in [12] as a way of studying the extremal properties of $\mathcal{F}$-free hypergraphs that goes well beyond just the determination of $\pi(\mathcal{F})$. In particular, we proved that $\Omega(F)$ is completely determined by a left-continuous almost everywhere differentiable function $g(\mathcal{F}) : \mathrm{proj}\Omega(\mathcal{F}) \to [0,1]$, where

$$\mathrm{proj}\Omega(\mathcal{F}) = \{x : \exists y \in [0,1] \text{ such that } (x,y) \in \Omega(\mathcal{F})\},$$

and

$$g(\mathcal{F},x) = \max\{y : (x,y) \in \Omega(\mathcal{F})\}, \text{ for all } x \in \mathrm{proj}\Omega(\mathcal{F}).$$

Theorem 1.7 together with Theorem 1.12 yield the following result.

Theorem 1.13. *The set* $\mathrm{proj}\Omega(\mathcal{M}) = [0,1]$*, and* $g(\mathcal{M},x) \le 4/9$ *for all* $x \in [0,1]$*. Moreover,* $g(\mathcal{M},x) = 4/9$ *iff* $x \in \{5/6, 8/9\}$.

In words, Theorem 1.13 says that $\mathcal{M}$-free 3-graphs can have any possible shadow density but the edge density is maximized for exactly two values of the shadow densities. This is a much stronger property than $\xi(\mathcal{M}) = 2$.

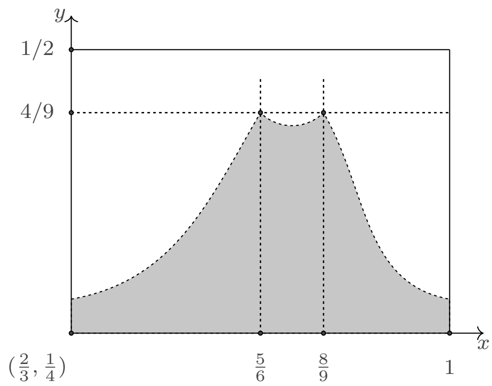


Figure 2: $g(\mathcal{M})$ has two global maximums by Theorem 1.13.

This paper is organized as follows. In Section 2 we will present some preliminary definitions and lemmas. In Section 3 we will prove Theorem 1.7, in Section 4 we will prove Theorem 1.12, and in Section 5 we will prove Theorem 1.13. We will include some remarks and open problems in Section 6.

2 Preliminaries

Let $r \ge 2$ and $\mathcal{H}$ be an r-graph. For every $v \in V(\mathcal{H})$ the degree of v in $\mathcal{H}$ is $d_{\mathcal{H}}(v) := |L_{\mathcal{H}}(v)|$, and the minimum degree of $\mathcal{H}$ is $\delta(\mathcal{H}) := \min\{d_{\mathcal{H}}(v) : v \in V(\mathcal{H})\}$. For $S \subset V(\mathcal{H})$, the neighborhood of S in $\mathcal{H}$ is

$$N_{\mathcal{H}}(S) := \{v \in V(\mathcal{H}) \setminus S : \exists E \in \mathcal{H} \text{ such that } \{v\} \cup S \subset E\}.$$

Two vertices $u, v \in V(\mathcal{H})$ are adjacent in $\mathcal{H}$ if $u \in N_{\mathcal{H}}(v)$. When it is clear from context we will omit the subscript $\mathcal{H}$ in the notations above.

Let $V(\mathcal{H}) = [n]$. For $x = (x_1, \ldots, x_n)$ define

$$p_{\mathcal{H}}(x) := \sum_{E \in \mathcal{H}} \prod_{i \in E} x_i.$$

The standard n-simplex is

$$\Delta^n := \left\{ x \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} x_i = 1 \text{ and } x_i \geq 0 \text{ for all } i \in [n+1] \right\}.$$

The lagrangian of $\mathcal{H}$ is

$$\lambda(\mathcal{H}) := \max \left\{ p_{\mathcal{H}}(x) : x \in \Delta^{n-1} \right\}.$$

Note that Δ^n is closed in $\mathbb{R}^{n+1}$ and $p_{\mathcal{H}}(x)$ is continuous, so $\lambda(\mathcal{H})$ is well-defined.

Recall that in Section 1 we defined the blow up of an r-graph $\mathcal{T}$. The next lemma gives a relationship between $\lambda(\mathcal{T})$ and the size of a blow up of $\mathcal{T}$.

Lemma 2.1. *Let $r \geq 2$ and $\mathcal{T}$ and $\mathcal{H}$ be two r-graphs. Suppose that $\mathcal{H}$ is a blow up of $\mathcal{T}$ with $v(\mathcal{H}) = n$. Then $|\mathcal{H}| \leq \lambda(\mathcal{T}) n^r$.*

Proof. Suppose that $|V(\mathcal{T})| = s$ and $\mathcal{H} = \mathcal{T}(t)$ for some $t = (t_1, \ldots, t_s)$. Then

$$|\mathcal{H}| = \sum_{E \in \mathcal{T}} \prod_{i \in E} t_i = n^r \sum_{E \in \mathcal{T}} \prod_{i \in E} \frac{t_i}{n} \leq \lambda(\mathcal{T}) n^r,$$

where the last inequality follows from the definition of $\lambda(\mathcal{T})$ and $\sum_{i \in [s]} t_i = n$. ∎

Given another r-graph F we say $f : V(F) \to V(\mathcal{H})$ is a homomorphism if $f(E) \in \mathcal{H}$ for all $E \in F$, i.e., f preserves edges. We say that $\mathcal{H}$ is F-hom-free if there is no homomorphism from F to $\mathcal{H}$. In other words, $\mathcal{H}$ is F-hom-free if and only if all blow ups of $\mathcal{H}$ are F-free. For a family $\mathcal{F}$ of r-graphs, $\mathcal{H}$ is $\mathcal{F}$-hom-free if it is F-hom-free for all $F \in \mathcal{F}$.

An r-graph F is 2-covered if every $\{u, v\} \subset V(F)$ is contained in some $E \in F$, and a family $\mathcal{F}$ is 2-covered if all $F \in \mathcal{F}$ are 2-covered. It is easy to see that if $\mathcal{F}$ is 2-covered, then $\mathcal{H}$ is $\mathcal{F}$-free if and only if it is $\mathcal{F}$-hom-free. Although $\mathcal{M}$ is not 2-covered, we still have a similar result.

Lemma 2.2. *A 3-graph $\mathcal{H}$ is $\mathcal{M}$-free if and only if it is $\mathcal{M}$-hom-free.*

Proof. It is clear that $\mathcal{M}$-hom-free implies $\mathcal{M}$-free. So it suffices to show that $\mathcal{M}$-free implies $\mathcal{M}$-hom-free.

Let $\mathcal{H}(t)$ be a blow up of $\mathcal{H}$, and suppose that $\mathcal{H}(t)$ contains a 3-graph $M \in \mathcal{M}$ as a subgraph. Since K_5^{3-} is 2-covered, $M \not\cong K_5^{3-}$. Similarly, since the core of every 3-graph in M_2 is 2-covered, $M \notin M_2$. Therefore, M is a 3-graph on six vertices such that $M \not\subset \mathcal{G}_n^1$ and $M \not\subset \mathcal{G}_n^2$ for all $n \geq 6$.

Let us assume that $V(M) = \{v_1, v_2, u_1, u_2, u_3, u_4\}$. Since $\mathcal{H}$ is M-free, we may assume that v_1, v_2 are contained in the same part of $\mathcal{H}(t)$, i.e. v_1 and v_2 are obtained from the same vertex in $\mathcal{H}$. Let M' denote the induced subgraph of M on $\{v_1, u_1, u_2, u_3, u_4\}$. Since $\mathcal{H}(t)$ is K_5^{3-}-free, $|M'| \leq 8$. A simple but crucial observation is that there are only two non-isomorphic 3-graphs on five vertices with exactly eight edges, and they are both contained in $\mathcal{G}_6^2$ as subgraphs. Therefore, $M' \subset \mathcal{G}_6^2$, and hence M is a subgraph of some blow up of $\mathcal{G}_6^2$, which implies that $M \subset \mathcal{G}_n^2$ for some n, a contradiction. ∎

3 Turán number of $\mathcal{M}$

In this section, we will prove Theorem 1.7. The first subsection contains some technical lemmas and calculations needed in the proof.

3.1 Lagrangian of some 3-graphs

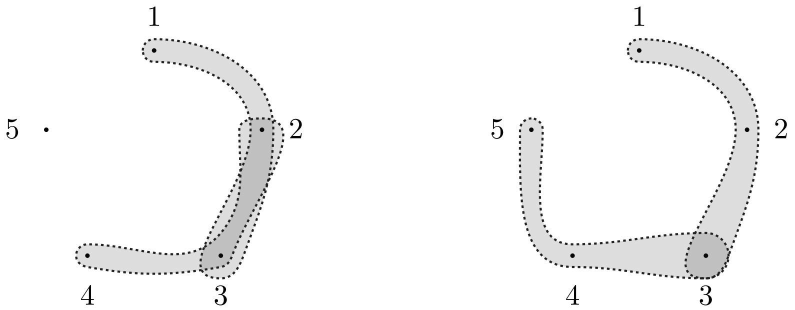
Lemma 3.1. *Suppose that $\mathcal{T}$ is a 3-graph with at most four vertices. Then $\lambda(\mathcal{T}) \le 1/16$.*

Proof. Without loss of generality we may assume that $v(\mathcal{T}) = 4$ and $|\mathcal{T}| = 4$, i.e., $\mathcal{T} \cong K_4^3$. It is easy to see that

$$p_{K_4^3}(x) = x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4 \le 4(1/4)^3 = 1/16.$$

Therefore, $\lambda(\mathcal{T}) \le 1/16$. ∎

Lemma 3.2. *Suppose that $\mathcal{T}$ is a 3-graph on five vertices with at most eight edges. Then $\lambda(\mathcal{T}) < 0.067277$.*



(a) The 3-graph on $\{1,2,3,4,5\}$ with missing edges $\{123, 234\}$.

(b) The 3-graph on $\{1,2,3,4,5\}$ with missing edges $\{123, 345\}$.

Figure 3: Two non-isomorphic 3-graphs on 5 vertices with 8 edges.

Proof. We may assume that $V(\mathcal{T}) = [5]$ and $|\mathcal{T}| = 8$ since adding edges into $\mathcal{T}$ will not decrease $\lambda(\mathcal{T})$. Then there are only two cases for $\mathcal{T}$, the missing edges in $\mathcal{T}$ are either $\{123, 234\}$ or $\{123, 345\}$.

Case 1: The missing edges in $\mathcal{T}$ are $\{123, 234\}$.
Then define

$$\mathcal{L}_1(x, \mu_1) = \sum_{ijk \subset [5]} x_ix_jx_k - (x_1x_2x_3 + x_2x_3x_4) - \mu_1 \left(\sum_{i=1}^{5} x_i - 1 \right).$$

Suppose that $p_{\mathcal{T}}(x)$ attains its maximum at the point $y = (y_1, y_2, y_3, y_4, y_5)$. Then $y_i > 0$ for all $i \in [5]$, since otherwise by Lemma 3.1, $p_{\mathcal{T}}(y) \le 1/16 = 0.0625$ and we are done. By the Lagrange multiplier method (e.g. see [6]),

$$\left.\frac{\partial \mathcal{L}_1}{\partial x_i}\right|_y = 0 \text{ for all } 1 \le i \le 5, \text{ and } \left.\frac{\partial \mathcal{L}_1}{\partial \mu_1}\right|_y = 0,$$

which implies

$$y_2y_4 + y_2y_5 + y_3y_4 + y_3y_5 + y_4y_5 - \mu_1 = 0, \tag{1}$$
$$y_1y_4 + y_1y_5 + y_3y_5 + y_4y_5 - \mu_1 = 0, \tag{2}$$
$$y_1y_4 + y_1y_5 + y_2y_5 + y_4y_5 - \mu_1 = 0, \tag{3}$$
$$y_1y_2 + y_1y_3 + y_1y_5 + y_2y_5 + y_3y_5 - \mu_1 = 0, \tag{4}$$
$$y_1y_2 + y_1y_3 + y_1y_4 + y_2y_3 + y_2y_4 + y_3y_4 - \mu_1 = 0, \tag{5}$$
$$\sum_{i \in [5]} y_i = 1. \tag{6}$$

(1) and (4) give $(y_1 - y_4)(y_2 + y_3 + y_5) = 0$, (2) and (3) give $(y_2 - y_3)y_5 = 0$, and it follows from $y_i > 0$ that $y_1 = y_4$ and $y_2 = y_3$. Let $a = y_1 = y_4$ and $b = y_2 = y_3$ and $c = y_5$. Then (1) $\sim$ (6) give

$$2ab + 2bc + ac - \mu_1 = 0 \tag{7}$$
$$a^2 + 2ac + bc - \mu_1 = 0 \tag{8}$$
$$a^2 + b^2 + 4ab - \mu_1 = 0 \tag{9}$$
$$2a + 2b + c = 1 \tag{10}$$

Substituting $c = 1 - (2a + 2b)$ and $\mu_1 = a^2 + b^2 + 4ab$ into (7) and (8) we obtain

$$3a^2 + 5b^2 + 8ab - a - 2b = 0 \tag{11}$$
$$4a^2 + 3b^2 + 10ab - 2a - b = 0 \tag{12}$$

Equations (11) and (12) give

$$3(3a^2 + 5b^2 + 8ab - a - 2b) - 5(4a^2 + 3b^2 + 10ab - 2a - b) = 0,$$

which implies

$$b = \frac{7a - 11a^2}{26a + 1}. \tag{13}$$

Substituting (13) into (11) we obtain

$$\frac{5a\left(69a^3 + 130a^2 - 18a - 3\right)}{(26a + 1)^2} = 0. \tag{14}$$

Equation (14) has exactly four solutions, and with the aid of a computer we can find that the roots are $\{0.2186380 \pm 10^{-6}, 0, -0.0992592 \pm 10^{-6}, -2.003437 \pm 10^{-6}\}$. Since $0 < a < 1$, $a = 0.2186380 \pm 10^{-6}$, and from (13), (10), and (9) we can obtain the values of b, c, and μ_1. Therefore,

$$\begin{aligned} y_1 = y_4 &= 0.218638 \pm 10^{-6}, \\ y_2 = y_3 &= 0.150292 \pm 10^{-6}, \\ y_5 &= 0.262141 \pm 10^{-6}, \\ \mu_1 &= 0.201828 \pm 10^{-6}, \\ p_{\mathcal{T}}(y) &= 0.067276 \pm 10^{-6}. \end{aligned}$$

Case 2: The missing edges are $\{123, 345\}$.
Then define

$$\mathcal{L}_2(x, \mu_2) = \sum_{ijk \subset [5]} x_i x_j x_k - (x_1 x_2 x_3 + x_3 x_4 x_5) - \mu_2 \left(\sum_{i=1}^{5} x_i - 1 \right).$$

Suppose that $p_{\mathcal{T}}(x)$ attains its maximum at point $y = (y_1, y_2, y_3, y_4, y_5)$. Then by the Lagrange multiplier method,

$$\left.\frac{\partial \mathcal{L}_2}{\partial x_i}\right|_y = 0 \text{ for all } 1 \le i \le 5, \text{ and } \left.\frac{\partial \mathcal{L}_2}{\partial \mu_2}\right|_y = 0.$$

Therefore,

$$\begin{aligned}
y_2y_4 + y_2y_5 + y_3y_4 + y_3y_5 + y_4y_5 - \mu_2 &= 0, \\
y_1y_4 + y_1y_5 + y_3y_4 + y_3y_5 + y_4y_5 - \mu_2 &= 0, \\
y_1y_4 + y_1y_5 + y_2y_4 + y_2y_5 - \mu_2 &= 0, \\
y_1y_2 + y_1y_3 + y_1y_5 + y_2y_3 + y_2y_5 - \mu_2 &= 0, \\
y_1y_2 + y_1y_3 + y_1y_4 + y_2y_3 + y_2y_4 - \mu_2 &= 0, \\
\sum_{i \in [5]} y_i &= 1.
\end{aligned}$$

Similar to Case 1, we obtain

$$\begin{aligned}
y_1 = y_2 = y_4 = y_5 &= 2/9, \\
y_3 &= 1/9, \\
\mu_2 &= 16/81, \\
p_{\mathcal{T}}(y) &= 16/243 = 0.0658436 \pm 10^{-6}.
\end{aligned}$$

Therefore, $\lambda(\mathcal{T}) < 0.067277$. ∎

Lemma 3.3. $\lambda(\mathcal{G}_6^2) \leq 2/27$.

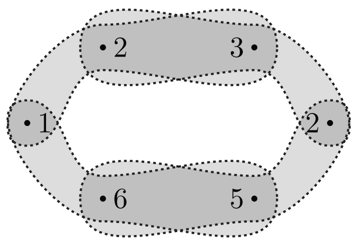


Figure 4: The 3-graph $\mathcal{G}_6^2$ with vertex set $[6]$ and missing edges $\{123, 234, 456, 561\}$.

Proof. Define

$$\mathcal{L}_3(x, \mu_3) = \sum_{ijk \subset [6]} x_ix_jx_k - (x_1x_2x_3 + x_2x_3x_4 + x_4x_5x_6 + x_5x_6x_1) - \mu_3 \left(\sum_{i \in [6]} x_i - 1 \right).$$

Suppose that $p_{\mathcal{G}_6^2}(x)$ attains its maximum at point $y = (y_1, y_2, y_3, y_4, y_5, y_6)$. Then by the Lagrange multiplier method,

$$\left.\frac{\partial \mathcal{L}_3}{\partial x_i}\right|_y = 0 \text{ for all } 1 \leq i \leq 6, \text{ and } \left.\frac{\partial \mathcal{L}_3}{\partial \mu_3}\right|_y = 0.$$

Therefore,

$$\begin{aligned}
y_4(y_2+y_3+y_5+y_6)+(y_2+y_3)(y_5+y_6)-\mu_3&=0,\\
y_1(y_2+y_3+y_5+y_6)+(y_2+y_3)(y_5+y_6)-\mu_3&=0,\\
\sum_{ij\subset\{1,4,5,6\}} y_iy_j+y_3(y_5+y_6)-\mu_3&=0,\\
\sum_{ij\subset\{1,4,5,6\}} y_iy_j+y_2(y_5+y_6)-\mu_3&=0,\\
\sum_{ij\subset\{1,2,3,4\}} y_iy_j+y_6(y_1+y_2)-\mu_3&=0,\\
\sum_{ij\subset\{1,2,3,4\}} y_iy_j+y_5(y_1+y_2)-\mu_3&=0,\\
\sum_{i\in[6]} y_i&=1.
\end{aligned}$$

Similar to Lemma 3.2, we obtain

$$\begin{aligned}
y_1=y_2=y_3=y_4=y_5=y_6&=1/6,\\
\mu_3&=2/9,\\
p_{\mathcal{G}_6^2}(y)&=2/27.
\end{aligned}$$

Therefore, $\lambda(\mathcal{G}_6^2)\le 2/27$. ∎

Notice that our proof of Lemma 3.3 actually shows that $p_{\mathcal{G}_6^2}(x)$ attains its maximum on Δ^5 only at $(1/6,\dots,1/6)$. This property will be used later in the proof of Claim 4.16.

An r-graph $\mathcal{T}$ is a star if all edges in $\mathcal{T}$ contain a fixed vertex v, which is called a center of $\mathcal{T}$.

Lemma 3.4. *Let $\mathcal{T}$ be a 2-covered 3-graph on $k\ge 7$ vertices. Suppose that $\tau(\mathcal{T}[S])\le 1$ for all sets $S\subset V(\mathcal{T})$ with $|S|=7$. Then $\mathcal{T}$ is a star.*

Proof. Suppose that $\mathcal{T}$ is not a star. Then for every vertex v in $\mathcal{T}$ there exists an edge E_v in $\mathcal{T}$ that does not contain v.

First notice that $\mathcal{T}$ cannot contain two disjoint edges. Therefore, $\mathcal{T}$ is intersecting. Suppose that $\mathcal{T}$ contains two edges $E_1=\{u,v_1,v_2\}$ and $E_2=\{u,w_1,w_2\}$, where $\{v_1,v_2\}\cap\{w_1,w_2\}=\emptyset$. Let $E_3\in\mathcal{T}$ be an edge that does not contain u. Since $\mathcal{T}$ is intersecting, we may assume that $v_1,w_1\in E_3$. Then, we have $|E_1\cup E_2\cup E_3|\le 6$, and $\tau(\{E_1,E_2,E_3\})=2$, a contradiction. Therefore, we may assume that the intersection of every two edges in $\mathcal{T}$ is two. Let $E_1=\{u,v,w_1\}$ and $E_2=\{u,v,w_2\}$ be two edges in $\mathcal{T}$. By assumption there exists an edge $E_3\in\mathcal{T}$ that does not contain u and, hence, we have $E_3=\{v,w_1,w_2\}$. Similarly there exists $E_4\in\mathcal{T}$ that does not contain v and, hence, we have $E_4=\{u,w_1,w_2\}$. Then, we have $|E_1\cup E_2\cup E_3\cup E_4|=4$, and $\tau(\{E_1,E_2,E_3,E_4\})=2$, a contradiction. ∎

3.2 Proof of Theorem 1.7

In this section we complete the proof of Theorem 1.7.

For $v\in V(\mathcal{H})$ and $E\in\mathcal{H}$, $\mathcal{H}-v$ is obtained by removing v and all edges containing v from $\mathcal{H}$, and $\mathcal{H}-E$ is obtained by removing E from $\mathcal{H}$ and keeping $V(\mathcal{H})$ unchanged.

Definition 3.5 (Equivalence classes)**.** *Let $\mathcal{H}$ be an r-graph and u,v be two non-adjacent vertices in $\mathcal{H}$. Then u and v are equivalent if $L(u)=L(v)$, otherwise they are non-equivalent. If u and v are equivalent, then we write $u\sim v$. Let C_v denote the equivalence class of v.*

Algorithm 1 (Symmetrization without cleaning) Let $\mathcal{H}$ be an r-graph. We perform the following operation as long as there are two non-adjacent non-equivalent vertices in $\mathcal{H}$. Let u, v be two such vertices with $d(u) \geq d(v)$. Then we delete all vertices in C_v and duplicate u using $|C_v|$ vertices and still label these new vertices with labels in C_v. Another way to view this operation is that we remove all edges in $\mathcal{H}_{i-1}$ that have nonempty intersection with C_v and for every $E \in \mathcal{H}_{i-1}$ with $u \in E$ we add $E - \{u\} \cup \{v'\}$ for all $v' \in C_v$ into $\mathcal{H}_{i-1}$. We terminate the process when there is no non-adjacent non-equivalent pair.

Note that the number of equivalence classes in $\mathcal{H}$ strictly decrease after each step that can be performed, so Algorithm 1 always terminates. On the other hand, since symmetrization only deletes and duplicates vertices, by Lemma 2.2, Algorithm 1 preserves the $\mathcal{M}$-freeness of $\mathcal{H}$. The following lemma is immediate from the definition.

Lemma 3.6. *Let $\mathcal{H}_t$ be the 3-graph obtained from $\mathcal{H}$ by applying Algorithm 1, and let $T \subset V(\mathcal{H})$ such that T contains exactly one vertex from each equivalence class of $\mathcal{H}_t$. Then,*

(a) $|\mathcal{H}_t| \geq |\mathcal{H}|$.

(b) $\mathcal{H}_t[T]$ is 2-covered and $\mathcal{H}_t$ is a blow-up of $\mathcal{H}_t[T]$.

Now we are ready to finish the proof of Theorem 1.7.

Proof of Theorem 1.7. Let $\mathcal{H}$ be an $\mathcal{M}$-free 3-graph on n vertices. Apply Algorithm 1 to $\mathcal{H}$ and let $\mathcal{H}_t$ denote the resulting 3-graph. Let $T \subset V(\mathcal{H})$ such that T contains exactly one vertex from each equivalent class in $\mathcal{H}_t$, and let $\mathcal{T} = \mathcal{H}_t[T]$. By Lemma 3.6, in order to prove $|\mathcal{H}| \leq 2n^3/27$, it suffices to show $|\mathcal{H}_t| \leq 2n^3/27$. Since $\mathcal{H}_t$ is a blow up of $\mathcal{T}$, by Lemma 2.1, it suffices to show that $\lambda(\mathcal{T}) \leq 2/27$. Next, we will consider two cases depending on the size of T: either $|T| \geq 7$ or $|T| \leq 6$.

Case 1: $|T| \geq 7$.
Since $\mathcal{T}$ is 2-covered and it is M_2-free, $\tau(\mathcal{T}[S]) \leq 1$ for all $S \subset T$ with $|S| = 7$, and it follows from Lemma 3.4 that $\mathcal{T}$ is a star.

Let us calculate $\lambda(\mathcal{T})$. We may assume that $V(\mathcal{T}) = [s]$ for some integer s and 1 is the center of $\mathcal{T}$. Then,

$$p_{\mathcal{T}}(x) \leq x_1 \left(\sum_{\{i,j\} \subset [s] \setminus \{1\}} x_i x_j \right) \leq \frac{s-2}{2(s-1)} x_1 (1-x_1)^2 < \frac{1}{2} x_1 (1-x_1)^2 \leq \frac{2}{27},$$

which implies that $\lambda(\mathcal{T}) < 2/27$.

Case 2: $|T| \leq 6$.
If $|T| \leq 5$, then Lemmas 3.1 and 3.2 imply that $\lambda(\mathcal{T}) < 0.67277$. So we may assume that $|T| = 6$.

Since $\mathcal{H}_t$ does not contain any member in M_3 as a subgraph, either $\mathcal{T} \subset \mathcal{G}_n^1$ or $\mathcal{T} \subset \mathcal{G}_n^2$ for some $n \geq 6$. Lemma 3.6 implies that $\mathcal{T}$ is 2-covered, therefore, either $\mathcal{T}$ is a star or $\mathcal{T} \cong \mathcal{G}_6^2$. The former case has been handled by Case 1, so we may assume that $\mathcal{T} \cong \mathcal{G}_6^2$, and it follows from Lemma 3.3 that $\lambda(\mathcal{T}) = \lambda(\mathcal{G}_6^2) \leq 2/27$. ∎

4 Stability of $\mathcal{M}$

In this section we will prove Theorem 1.12. First we present an algorithm and some lemmas that will be used in the proof.

4.1 Symmetrization

Let $0 \le \alpha \le 1$ and $\mathcal{H}$ be a 3-graph. Then $\mathcal{H}$ is α-dense if $\delta(\mathcal{H}) \ge \alpha\binom{v(\mathcal{H})-1}{2}$. Let $(V, \prec_V)$ be a poset on V with relation $\prec_V$. For $S \subset V$ the induced poset of $(V, \prec_V)$ on S is denoted by $(S, \prec_V)$.

Algorithm 2 (Symmetrization and cleaning with threshold α).
Input: A 3-graph $\mathcal{H}$.
Operation:

- **Initial step**: If $\delta(\mathcal{H}) \ge \alpha\binom{v(\mathcal{H})-1}{2}$, then let $\mathcal{H}_0 = \mathcal{H}$ and $V_0 = V(\mathcal{H})$. Otherwise, we keep deleting vertices with the minimum degree one by one until the remaining 3-graph $\mathcal{H}_0$ is either empty or $\delta(\mathcal{H}_0) \ge \alpha\binom{v(\mathcal{H}_0)-1}{2}$. Let Z_0 be the set of deleted vertices during this process so that $V_0 := V(\mathcal{H}_0) = V(\mathcal{H}) - Z_0$.

Let $(V_0, \prec_{V_0})$ be the poset with V_0 itself an antichain, i.e., there is no relation between any two vertices in V_0.

Suppose we are at the i-th step for some $i \ge 1$. We terminate the algorithm if either

(a) $\mathcal{H}_{i-1} = \emptyset$ or

(b) $\delta(\mathcal{H}_{i-1}) \ge \alpha\binom{v(\mathcal{H}_{i-1})-1}{2}$ and there is no pair of non-adjacent non-equivalent vertices.

Otherwise, we iterate the following two operations.

- **Iteration**. For $i \ge 1$, Step 1 together with Step 2 will transform $\mathcal{H}_{i-1}$ into $\mathcal{H}_i$, and we will iterate Step 1 and Step 2 until we get a 3-graph $\mathcal{H}_t$ for some t such that either $\mathcal{H}_t$ is empty or $\delta(\mathcal{H}_t) \ge \alpha\binom{v(\mathcal{H}_t)-1}{2}$ and there is no pair of non-adjacent non-equivalent vertices in $\mathcal{H}_t$.

  Step 1 (Symmetrization): If $\mathcal{H}_{i-1}$ contains no pair of non-adjacent non-equivalent vertices, then let $\mathcal{G}_i = \mathcal{H}_{i-1}$ and go to Step 2. Otherwise, choose two non-adjacent non-equivalent vertices $u, v \in V(\mathcal{H}_{i-1})$ and assume that $d(u) \ge d(v)$. Delete all vertices in C_v and add $|C_v|$ new vertices into C_u by duplicating u and label these new vertices with labels in C_v, which is the same as what we did in Algorithm 1. Let $\mathcal{G}_i$ denote the resulting r-graph, and update the poset $(V_{i-1}, \prec_{V_{i-1}})$ by adding the following relations: $v' \prec u'$ for all $v' \in C_v$ and all $u' \in C_u$. This new poset is well-defined as one will see from the following operations that once two equivalence classes are merged they will never be split.

  Step 2 (Cleaning): If $\delta(\mathcal{G}_i) \ge \alpha\binom{v(\mathcal{G}_i)-1}{2}$, then let $\mathcal{H}_i = \mathcal{G}_i$ and $(V_i, \prec_{V_i}) = (V_{i-1}, \prec_{V_{i-1}})$. Otherwise let $\mathcal{L} = \mathcal{G}_i$ and repeat Steps 2.1 and 2.2.

  Step 2.1: Let $B = \{a \in V(\mathcal{L}) : d_{\mathcal{L}}(a) = \delta(\mathcal{L})\}$ and choose a minimal element $z \in (B, \prec_{V_{i-1}})$. Let $\mathcal{L} = \mathcal{L} - z$, $V_{i-1} = V_{i-1} \setminus \{z\}$, and $(V_{i-1}, \prec_{V_{i-1}}) = (V_{i-1} \setminus \{z\}, \prec_{V_{i-1}})$.

  Step 2.2: If $\delta(\mathcal{L}) \ge \alpha\binom{v(\mathcal{L})-1}{2}$ or $\mathcal{L} = \emptyset$, then stop. Otherwise, go to Step 2.1.

  Let $\mathcal{H}_i = \mathcal{L}$ and $(V_i, \prec_{V_i}) = (V_{i-1}, \prec_{V_{i-1}})$. Let Z_i denote the set of vertices removed by Step 2.1 so that $\mathcal{H}_i = \mathcal{G}_i - Z_i$.

Output: A 3-graph $\mathcal{H}_t$ for some t such that either $\mathcal{H}_t$ is empty or $\delta(\mathcal{H}_t) \ge \alpha\binom{v(\mathcal{H}_t)-1}{2}$ and there is no pair of non-adjacent non-equivalent vertices in $\mathcal{H}_t$.

Let $\epsilon > 0$ be sufficiently small and n be sufficiently large (say $n > 1/\epsilon$). Let $\mathcal{H}$ be an $\mathcal{M}$-free 3-graph on n vertices with $|\mathcal{H}| \ge 2n^3/27 - \epsilon n^3$. Apply Algorithm 2 to $\mathcal{H}$ with threshold $4/9 - 3\epsilon^{1/2}$ and suppose that it stops at the t-th step. Let $\mathcal{H}_t$ denote the resulting

3-graph and $W = V(\mathcal{H}_t)$ and $\tilde{n} = |W|$. For $0 \le i \le t$ let $\widetilde{\mathcal{H}}_i = \mathcal{H}_i[W]$ and $\widetilde{\mathcal{G}}_i = \mathcal{G}_i[W]$. Note that $\widetilde{\mathcal{H}}_0 = \mathcal{H}[W]$ and $\widetilde{\mathcal{G}}_0 = \mathcal{G}[W]$, and we will omit the subscript 0 if there is no cause for confusion. Let $Z = \bigcup_{i=0}^{t} Z_i$ be the set of vertices in $\mathcal{H}$ that were removed by Algorithm 2. In the rest of the proof we will focus on $\widetilde{\mathcal{H}}_i$ and $\widetilde{\mathcal{G}}_i$. Notice from Algorithm 2 that $\mathcal{H}_i = \mathcal{G}_i - Z_i$ and $Z_i \subset V(\mathcal{H}) \setminus W$, therefore, $\widetilde{\mathcal{H}}_i = \widetilde{\mathcal{G}}_i$ for all $1 \le i \le t$.

$$\begin{tikzcd}
\mathcal{H} \arrow[r, "-Z_0", "\supseteq"'] & \mathcal{H}_0 \arrow[r, "\text{Sym.}"] & \mathcal{G}_1 \arrow[r, "-Z_1"] & \mathcal{H}_1 \arrow[r, "\text{Sym.}"] & \cdots \arrow[r, "-Z_{t-1}"] & \mathcal{H}_{t-1} \arrow[r, "\text{Sym.}"] & \mathcal{G}_t \arrow[r, "-Z_t"] & \mathcal{H}_t \\
& \widetilde{\mathcal{H}}_0 \arrow[ul, "\subseteq"] \arrow[u, "\subseteq"'] & & \widetilde{\mathcal{H}}_1 \arrow[ll] \arrow[u, "\subseteq"'] & \cdots \arrow[l] & \widetilde{\mathcal{H}}_{t-1} \arrow[l] \arrow[u, "\subseteq"'] & & \widetilde{\mathcal{H}}_t \arrow[ll] \arrow[u, "="]
\end{tikzcd}$$

Figure 5: The first line contains the 3-graphs produced by Algorithm 2 and the second line contains the corresponding induced 3-graphs on W.

Lemma 4.1. *For every $i \in [t]$ either $\widetilde{\mathcal{H}}_{i-1} = \widetilde{\mathcal{H}}_i$ or there exist two nonempty equivalence classes $V_i \subset W$ and $U_i \subset W$ in $\widetilde{\mathcal{H}}_{i-1}$ such that $\widetilde{\mathcal{H}}_i$ is obtained from $\widetilde{\mathcal{H}}_{i-1}$ by deleting all vertices in V_i and adding $|V_i|$ new vertices by duplicating some vertex in U_i.*

Proof. Fix $1 \le i \le t$ and suppose that in forming $\mathcal{G}_i$ from $\mathcal{H}_{i-1}$ in Algorithm 2 we deleted all vertices in C_v and added $|C_v|$ new vertices by duplicating some $u \in C_u$, where C_v (resp. C_u) is the equivalence class of $v \in V(\mathcal{H}_{i-1})$ (resp. $u \in V(\mathcal{H}_{i-1})$) in $\mathcal{H}_{i-1}$. Let $\widehat{C}_u = C_v \cup C_u$ and notice that for every $i \le j \le t$ the set $\widehat{C}_u \cap V(\mathcal{G}_j)$ (resp. $\widehat{C}_u \cap V(\mathcal{H}_j)$) is an equivalence class in $\mathcal{G}_j$ (resp. $\mathcal{H}_j$).

If $C_v \cap W = \emptyset$, then $\widetilde{\mathcal{H}}_{i-1} = \widetilde{\mathcal{H}}_i$ and we are done. So we may assume that $C_v \cap W \neq \emptyset$.

First, we claim that $C_u \subset W$. Indeed, suppose that there exists $u' \in C_u \setminus W$. Then it means that u' was removed at the j-th step for some $i \le j \le t$. Since all $v' \in C_v$ satisfies $v' \prec_{V_k} u'$ and $d_{\mathcal{G}_k}(v') = d_{\mathcal{G}_k}(u')$ for all $i \le k \le j$, according to Algorithm 2 all vertices in C_v must be removed before u' was removed, which implies that $C_v \cap W = \emptyset$, a contradiction. Therefore, $C_u \subset W$.

Let $V_i = C_v \cap W$ and $U_i = C_u$ and note that none of them is empty. Since C_v and C_u are equivalence classes in $\mathcal{H}_{i-1}$, V_i and U_i are equivalence classes in $\widetilde{\mathcal{H}}_{i-1}$. According to Algorithm 2, $\widetilde{\mathcal{H}}_i$ is obtained from $\widetilde{\mathcal{H}}_{i-1}$ by deleting all vertices in V_i and adding $|V_i|$ new vertices by duplicating some vertex in U_i. ∎

The next lemma shows that $|Z|$ is small.

Lemma 4.2. *$|Z| \le 3\epsilon^{1/2} n$, and hence $\tilde{n} \ge n - 3\epsilon^{1/2} n$.*

Proof. Let $p = |Z|$. Then

$$\begin{aligned}
|\mathcal{H}_t| &\ge |\mathcal{H}| - \left(\frac{4}{9} - 3\epsilon^{1/2}\right) \sum_{i=1}^{p} \binom{n-i}{2} \\
&\ge \left(\frac{4}{9} - 6\epsilon\right) \binom{n}{3} - \left(\frac{4}{9} - 3\epsilon^{1/2}\right) \left(\binom{n}{3} - \binom{n-p}{3}\right) \\
&= \left(\frac{4}{9} - 3\epsilon^{1/2}\right) \binom{n-p}{3} + \left(3\epsilon^{1/2} - 6\epsilon\right) \binom{n}{3}.
\end{aligned} \tag{15}$$

First, we claim that $p < n/2$. Otherwise, (15) gives

$$\begin{aligned}|\mathcal{H}_t| &\geq \left(\frac{4}{9} - 3\epsilon^{1/2}\right)\binom{n-p}{3} + \left(3\epsilon^{1/2} - 6\epsilon\right)\binom{n}{3} \\ &= \frac{4}{9}\binom{n-p}{3} + \left(3\epsilon^{1/2} - 6\epsilon\right)\binom{n}{3} - 3\epsilon^{1/2}\binom{n/2}{3} \\ &> \frac{4}{9}\binom{n-p}{3} + \epsilon^{1/2}\binom{n}{3} > \frac{2}{27}(n-p)^3,\end{aligned}$$

which, by Lemma 3.6 and Theorem 1.7, is a contradiction. Here we used the assumption that ϵ is sufficiently small and $n > 1/\epsilon$ is sufficiently large.

Since $n - p > n/2$ is still sufficiently large, by Lemma 3.6 and Theorem 1.7,

$$|\mathcal{H}_t| \leq \frac{2}{27}(n-p)^3 < \left(\frac{4}{9} + \epsilon\right)\binom{n-p}{3}. \tag{16}$$

Equations (15) and (16) give

$$\left(\frac{4}{9} + \epsilon\right)\binom{n-p}{3} > \left(\frac{4}{9} - 6\epsilon\right)\binom{n}{3} - \left(\frac{4}{9} - 3\epsilon^{1/2}\right)\left(\binom{n}{3} - \binom{n-p}{3}\right).$$

It follows that

$$(3\epsilon^{1/2} + \epsilon)\binom{n-p}{3} > (3\epsilon^{1/2} - 6\epsilon)\binom{n}{3},$$

and hence

$$\left(\frac{n-p}{n}\right)^3 > \frac{\binom{n-p}{3}}{\binom{n}{3}} > \frac{3\epsilon^{1/2} - 6\epsilon}{3\epsilon^{1/2} + \epsilon} > 1 - 3\epsilon^{1/2},$$

which implies that $p < 3\epsilon^{1/2}n$ and $\tilde{n} = n - p > n - 3\epsilon^{1/2}n$. ∎

Since $\widetilde{\mathcal{H}}_t = \mathcal{H}_t$ is $(4/9 - 3\epsilon^{1/2})$-dense, $\delta(\widetilde{\mathcal{H}}_t) \geq (4/9 - 3\epsilon^{1/2})\binom{|W|-1}{2}$. The next lemma shows that $\delta(\widetilde{\mathcal{H}}_i)$ is also large for $0 \leq i \leq t-1$.

Lemma 4.3. *For all* $0 \leq i \leq t$,

$$\delta(\widetilde{\mathcal{H}}_i) > \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2}$$

and, in particular,

$$|\widetilde{\mathcal{H}}_i| > \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}}{3}.$$

Proof. It follows from Algorithm 2 that

$$\delta(\mathcal{H}_i) \geq \left(\frac{4}{9} - 3\epsilon^{1/2}\right)\binom{v(\mathcal{H}_i)-1}{2}.$$

By Lemma 4.2, for each $w \in W$ there are at most $3\epsilon^{1/2}n^2$ edges in $L_{\mathcal{G}_i}(w)$ that have nonempty intersection with $V(\mathcal{H}) \setminus W$. Therefore,

$$d_{\widetilde{\mathcal{H}}_i}(w) \geq \left(\frac{4}{9} - 3\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2} - 3\epsilon^{1/2}n^2 \overset{\text{Lemma 4.2}}{>} \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2},$$

and it follows that

$$|\widetilde{\mathcal{H}}_i| = \frac{1}{3}\sum_{w\in W} d_{\widetilde{\mathcal{H}}_i}(w) > \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}}{3}.$$

∎

Let $T \subset W$ such that T contains exactly one vertex in each equivalence class in $\mathcal{H}_t$ and $\mathcal{T} = \mathcal{H}_t[T]$. Lemma 3.6 implies that $\mathcal{T}$ is 2-covered and $\mathcal{H}_t$ is a blow up of $\mathcal{T}$. Our next lemma will show that either $\mathcal{T}$ is a (maximum) star or $\mathcal{T} \cong \mathcal{G}_6^2$. If $\mathcal{T}$ is a (maximum) star, then $\mathcal{H}_t$ is semibipartite and there are two parts in $\mathcal{H}_t$, one part is the blow up of the center of $\mathcal{T}$ and the other part is the union of all blow ups of non-center vertices of $\mathcal{T}$. If $\mathcal{T} \cong \mathcal{G}_6^2$, then $\mathcal{H}_t$ is $\mathcal{G}_6^2$-colorable and there are six parts in $\mathcal{H}_t$, each of them is a blow up of a vertex in $\mathcal{T}$.

Lemma 4.4. *Either $\mathcal{T}$ is a (maximum) star or $\mathcal{T} \cong \mathcal{G}_6^2$ and, in particular, $\mathcal{H}_t$ is either semibipartite or $\mathcal{G}_6^2$-colorable, Moreover, each part in $\mathcal{H}_t$ is a union of some (possibly just one) equivalence classes in $\mathcal{H}_t$.*

Proof. First we claim that $|T| \geq 6$. Indeed, suppose that $|T| \leq 5$. Then, Lemmas 3.1 and 3.2 imply that $\lambda(\mathcal{T}) < 0.067277$. It follows from Lemma 2.1 that $|\mathcal{H}_t| < 0.067277\tilde{n}^3 < (4/9 - 10\epsilon^{1/2})\binom{\tilde{n}}{3}$, which contradicts Lemma 4.3. Therefore, $|T| \geq 6$.

Suppose that $|T| \geq 7$. Since $\mathcal{T}$ is 2-covered and M_2-free, $\tau(\mathcal{T}[S]) \leq 1$ for all $S \subset T$ with $|S| = 7$. So by Lemma 3.4, $\mathcal{T}$ is a star, and hence $\mathcal{H}_t$ is semibipartite.

Suppose that $|T| = 6$. Since $\mathcal{H}$ is M_3-free, either $\mathcal{T} \subset \mathcal{G}_m^1$ or $\mathcal{T} \subset \mathcal{G}_m^2$ for some $m \geq 6$. On the other hand, $\mathcal{T}$ is 2-covered, therefore, either $\mathcal{T}$ is a star or $\mathcal{T} \cong \mathcal{G}_6^2$. In the former case, $\mathcal{H}_t$ is semibipartite, and in the later case, $\mathcal{H}_t$ is $\mathcal{G}_6^2$-colorable.

The later part of Lemma 4.4 is clear from Lemma 3.6. ∎

Notice that $\widetilde{\mathcal{H}}_t = \mathcal{H}_t$ and $\widetilde{\mathcal{H}}_0 = \mathcal{H}[W]$. In order to prove Theorem 1.12 it suffices to show that $\widetilde{\mathcal{H}}_0$ is either semibipartite or $\mathcal{G}_6^2$-colorable. We will proceed by backward induction on i with the base case from Lemma 4.4 and we will consider two cases in the following two subsections depending on the structure of $\mathcal{H}_t$.

4.2 Semibipartite

In this section we will prove the following statement.

Lemma 4.5. *Suppose that $\mathcal{H}_t$ is semibipartite and each part in $\mathcal{H}_t$ is a union of some equivalence classes in $\mathcal{H}_t$. Then $\widetilde{\mathcal{H}}_i$ is semibipartite and each part in $\widetilde{\mathcal{H}}_i$ is a union of some equivalence classes in $\widetilde{\mathcal{H}}_i$ for all $0 \leq i \leq t$. In particular, $\mathcal{H}[W] = \widetilde{\mathcal{H}}_0$ is semibipartite.*

Recall from Section 1 that the edge density of an r-graph $\mathcal{H}$ on n vertices is $d(\mathcal{H}) := |\mathcal{H}|/\binom{n}{r}$. For $r = 2$ Turán's theorem implies that if $\ell \geq 3$ is fixed and n is sufficiently large and $d(\mathcal{H}) > (\ell - 1)/\ell$, then $K_{\ell+1} \subset \mathcal{H}$. In our proof we will need the following results.

Theorem 4.6 (see Moon and Moser [14])**.** *Let $\ell \geq 3$ and $x \geq (\ell - 2)/(2(\ell - 1))$ and G be a graph on n vertices with at least xn^2 edges. Then, G contains at least*

$$f_\ell(n, x) := \frac{\Pi_{i=1}^{\ell-1}(2ix - (i-1))}{\ell!} n^\ell$$

copies of K_ℓ.

Theorem 4.7 (see Lovász [13])**.** *Let $\ell \geq 3$ and $x \geq 0$ and G be a graph on n vertices with at most xn^2 edges. Then, G contains at most*

$$h_\ell(n, x) := \frac{(2x)^{\ell/2}}{\ell!} n^\ell$$

copies of K_ℓ.

Now we prove Lemma 4.5.

Proof of Lemma 4.5. The proof is by backward induction on i and the base case is $i = t$ as $\widetilde{\mathcal{H}}_t = \mathcal{H}_t$. Now suppose that $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite with two parts A^{i+1} and B^{i+1} for some $0 \le i \le t-1$. Our goal is to show that $\widetilde{\mathcal{H}}_i$ is also semibipartite.

Claim 4.8.
$$\left| |A^{i+1}| - \frac{\tilde{n}}{3} \right| < 4\epsilon^{1/4}\tilde{n}, \text{ and } \left| |B^{i+1}| - \frac{2\tilde{n}}{3} \right| < 4\epsilon^{1/4}\tilde{n}.$$

Proof of Claim 4.8. Let $\beta = |B^{i+1}|$. Since $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite,
$$|\widetilde{\mathcal{H}}_{i+1}| \le (\tilde{n} - \beta)\binom{\beta}{2}.$$

On the other hand, by Lemma 4.3,
$$|\widetilde{\mathcal{H}}_{i+1}| \ge (4/9 - 10\epsilon^{1/2})\binom{\tilde{n}}{3}.$$

Therefore,
$$(4/9 - 10\epsilon^{1/2})\binom{\tilde{n}}{3} \le (\tilde{n} - \beta)\binom{\beta}{2},$$
which implies that $(2/3 - 4\epsilon^{1/4})\tilde{n} < \beta < (2/3 + 4\epsilon^{1/4})\tilde{n}$. ∎

Lemma 4.1 implies that either $\widetilde{\mathcal{H}}_i = \widetilde{\mathcal{H}}_{i+1}$ or there exists two equivalence classes V and U in $\widetilde{\mathcal{H}}_i$ such that $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing V to U. In the former case, there is nothing to prove, so we may assume that we are in the later case. Suppose that V is the equivalence class of $v \in W$ and U is the equivalence class of $u \in W$ (in $\widetilde{\mathcal{H}}_i$), i.e. $V = C_v$ and $U = C_u$. Let $L = A^{i+1} \setminus C_v$, $R = B^{i+1} \setminus C_v$ and $W' = W \setminus C_v$. Since $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing C_v to C_u and C_u is contained in either L or R, it follows that $L \neq \emptyset$ and $R \neq \emptyset$.

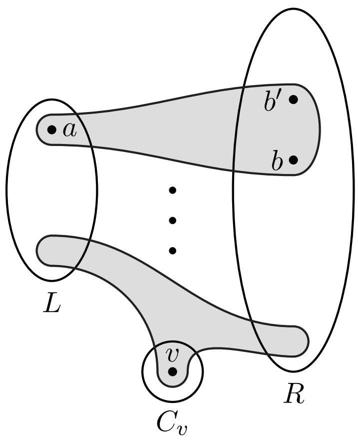


Figure 6: The 3-graph $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing C_v to some equivalence class C_u that contained in L or R.

Since C_v is an equivalence class in $\widetilde{\mathcal{H}}_i$, $L_{\widetilde{\mathcal{H}}_i}(v') = L_{\widetilde{\mathcal{H}}_i}(v)$ for all $v' \in C_v$, and hence we may just focus on v. Notice that in forming $\widetilde{\mathcal{H}}_{i+1}$ from $\widetilde{\mathcal{H}}_i$ we only delete and add edges that have nonempty intersection with C_v, so $\widetilde{\mathcal{H}}_i[W'] = \widetilde{\mathcal{H}}_{i+1}[W']$. Since $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite, it follows that $\widetilde{\mathcal{H}}_i[W']$ is semibipartite with two parts L and R.

If $N_{\widetilde{\mathcal{H}}_i}(v) \subset R$, then let $A^i = L \cup C_v$ and $B^i = B^{i+1} \setminus C_v$ and it is easy to see that $\widetilde{\mathcal{H}}_i$ is semibipartite with two parts A^i and B^i, and we are done. So we may assume that
$$N_{\widetilde{\mathcal{H}}_i}(v) \cap L \neq \emptyset. \qquad (*)$$

Our goal in the rest of the proof is to show that $L_{\widetilde{\mathcal{H}}_i}(v)$ is a bipartite graph with two parts L and R.

Claim 4.9. *$|N_{\widetilde{\mathcal{H}}_i}(v) \cap R| \geq \left(1/3 - 5\epsilon^{1/4}\right)\tilde{n}$. In particular, $|R| \geq \left(1/3 - 5\epsilon^{1/4}\right)\tilde{n}$.*

Proof of Claim 4.9. By Lemma 4.3,

$$\binom{|N_{\widetilde{\mathcal{H}}_i}(v)|}{2} \geq d_{\widetilde{\mathcal{H}}_i}(v) \geq \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2},$$

which implies that $|N_{\widetilde{\mathcal{H}}_i}(v)| \geq (2/3 - \epsilon^{1/2})\tilde{n}$. By Claim 4.8, $|L| \leq (1/3 + 4\epsilon^{1/4})\tilde{n}$, and hence

$$|N_{\widetilde{\mathcal{H}}_i}(v) \cap R| \geq \left(\frac{2}{3} - 15\epsilon^{1/2}\right)\tilde{n} - \left(\frac{1}{3} + 4\epsilon^{1/4}\right)\tilde{n} > \left(\frac{1}{3} - 5\epsilon^{1/4}\right)\tilde{n}.$$

∎

For every $w \in W$, let $N_L(w) = N_{\widetilde{\mathcal{H}}_i}(w) \cap L$, $N_R(w) = N_{\widetilde{\mathcal{H}}_i}(w) \cap R$, and $L(w) = L_{\widetilde{\mathcal{H}}_i}(w)$.

Claim 4.10. *Let $w \in L$ and $R' \subset R$ with $|R'| \geq c\tilde{n}$ for some constant $c > 0$. Then $d(L(w)[R']) > 1 - 40\epsilon^{1/4}/c^2$.*

Proof of Claim 4.10. It follows from $\widetilde{\mathcal{H}}_i[W'] = \widetilde{\mathcal{H}}_{i+1}[W']$ and $R' \subset R$ that $\binom{R'}{2} \setminus L_{\widetilde{\mathcal{H}}_i}(w)[R] = \binom{R'}{2} \setminus L_{\widetilde{\mathcal{H}}_{i+1}}(w)[R]$. On the other hand, since $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite, $L_{\widetilde{\mathcal{H}}_{i+1}}(w) \subset \binom{B^{i+1}}{2}$. Therefore, by Lemma 4.3 and Claim 4.8,

$$\begin{aligned}\left|\binom{R'}{2} \setminus L_{\widetilde{\mathcal{H}}_i}(w)[R]\right| = \left|\binom{R'}{2} \setminus L_{\widetilde{\mathcal{H}}_{i+1}}(w)[R]\right| &\leq \left|\binom{B^{i+1}}{2} \setminus L_{\widetilde{\mathcal{H}}_{i+1}}(w)\right| \\ &< \frac{1}{2}\left(\frac{2}{3} + 4\epsilon^{1/4}\right)^2\tilde{n}^2 - \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2} \\ &< 10\epsilon^{1/4}\tilde{n}^2,\end{aligned}$$

which implies that

$$d(L(w)[R']) = d(L_{\widetilde{\mathcal{H}}_i}(w)[R']) > 1 - \frac{10\epsilon^{1/4}\tilde{n}^2}{\binom{|R'|}{2}} > 1 - \frac{10\epsilon^{1/4}\tilde{n}^2}{c^2\tilde{n}^2/4} > 1 - 40\epsilon^{1/4}/c^2.$$

∎

Claim 4.11. *$L(v)[L] = \emptyset$.*

Proof of Claim 4.11. Suppose that there exists $w_1w_2 \in L(v)[L]$. For $j \in \{1,2\}$ let $L_R(w_j) = L_{\widetilde{\mathcal{H}}_i}(w_j)[R]$. We are going to show that there exists $K_5 \subset L_R(w_1) \cap L_R(w_2)$ and this will imply that $\widetilde{\mathcal{H}}_i$ contains a copy of a 3-graph in M_2 (see Figure 6), which is a contradiction.

For $j \in \{1,2\}$ letting $R' = R$ in Claim 4.10 we obtain

$$d\left(L_R(w_j)\right) > 1 - \frac{40\epsilon^{1/4}}{(|R|/\tilde{n})^2} \overset{\text{Claim 4.9}}{>} 1 - \frac{40\epsilon^{1/4}}{(1/3 - 5\epsilon^{1/4})^2} > 1 - 400\epsilon^{1/4}.$$

Therefore, $d\left(L_R(w_1) \cap L_R(w_2)\right) \geq 1 - 800\epsilon^{1/4}$, and by Turán's theorem, $K_5 \subset L_R(w_1) \cap L_R(w_2)$. We may assume that the vertex set of this K_5 is $\{w_3, w_4, w_5, w_6, w_7\}$. However, the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{w_1, \ldots, w_7\}$ is a copy of a 3-graph in M_2, a contradiction.
∎

Claim 4.12. *$|L(v)[R]| < 7000\epsilon^{1/4}\tilde{n}^2$.*

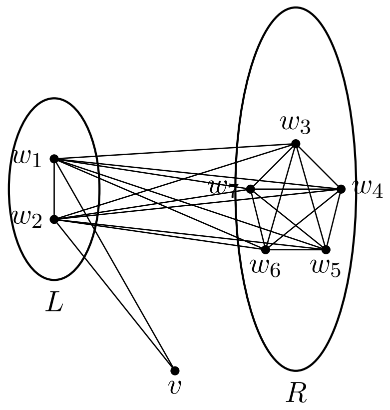


Figure 7: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{w_1, \ldots, w_7\}$ is a copy of a 3-graph in M_2, since both $w_1w_3w_4$ and $w_2w_5w_6$ are in $\widetilde{\mathcal{H}}_i$ and they are disjoint.

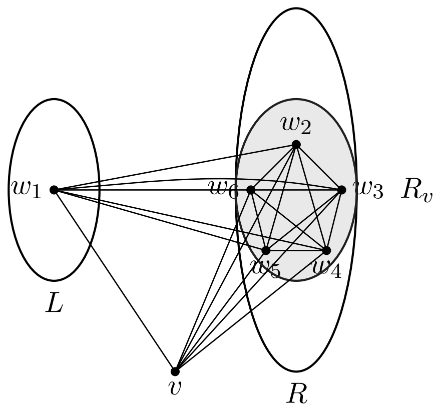


Figure 8: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_6\}$ is a copy of a 3-graph in M_2 since both vw_5w_6 and $w_1w_2w_3$ are in $\widetilde{\mathcal{H}}_i$ and they are disjoint.

Proof of Claim 4.12. Suppose that $|L(v)[R]| \geq 2000\epsilon^{1/4}\tilde{n}^2$. Let $R_v = N_{\widetilde{\mathcal{H}}_i}(v) \cap R$. Since by $(*)$, $N_L(v) \neq \emptyset$, choose $w_1 \in N_L(v)$ and let $L_{R,v}(w_1) = L_{\widetilde{\mathcal{H}}_i}(w_1)[R_v]$. Letting $R' = R_v$ in Claim 4.10 we obtain

$$|L_{R,v}(w_1)| = d(L_{R,v}(w_1))\binom{|R_v|}{2} > \left(1 - 400\epsilon^{1/4}\right)\binom{|R_v|}{2} > \frac{1 - 500\epsilon^{1/4}}{2}|R_v|^2.$$

Therefore, Theorem 4.6 implies that the number of copies of K_5 in $L_{R,v}(w_1)$ is at least

$$\begin{aligned} f_5(|R_v|, (1 - 500\epsilon^{1/4})/2) &> \frac{\prod_{i=1}^{4}\left(i(1 - 500\epsilon^{1/4}) - (i-1)\right)}{5!}|R_v|^5 \\ &> \frac{1 - 6000\epsilon^{1/4}}{5!}|R_v|^5. \end{aligned} \tag{17}$$

Suppose that there exists a copy of K_5 in $L_{R,v}(w_1)$, say on $\{w_2, w_3, w_4, w_5, w_6\}$, such that $w_5w_6 \in L(v)$. Then, the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_6\}$ would be a copy of a 3-graph in M_2, a contradiction. Therefore, no copy of K_5 in $L_{R,v}(w_1)$ contains any edge in $L(v)$. Then, by the assumption that $|L(v)[R]| \geq 7000\epsilon^{1/4}\tilde{n}^2$, there are at most

$$\frac{1}{2}|R_v|^2 - 7000\epsilon^{1/4}|R_v|^2 < \frac{1 - 7000\epsilon^{1/4}}{2}|R_v|^2$$

edges in $L_{R,v}(w_1)$ that are covered by some copy of K_5. Therefore, by Theorem 4.7, the number of copies of K_5 in $L_{R,v}(w_1)$ is at most

$$\begin{aligned} h_5(|R_v|, (1 - 7000\epsilon^{1/4})/2) &\leq \frac{(1 - 7000\epsilon^{1/4})^{5/2}}{5!}|R_v|^5 \\ &< \frac{1 - 7000\epsilon^{1/4}}{5!}|R_v|^5, \end{aligned}$$

which contradicts (17). ∎

For a graph G and $A, B \subset V(G)$ let

$$G[A, B] = \{uv \in G : u \in A \text{ and } v \in B\}.$$

Claim 4.12 implies that $|L(v) \setminus L(v)[L, R]| < 7000\epsilon^{1/4}\tilde{n}^2$, and we will show later that $L(v) = L(v)[L, R]$.

Claim 4.13. $|C_v| < 30000\epsilon^{1/4}|W|$.

Proof of Claim 4.13. Let $\alpha = |C_v|$ and notice that $V(L(v)) \subset L \cup R$ as C_v is an equivalence class in $\widetilde{\mathcal{H}}_i$. It follows from Claims 4.11, 4.12, and Lemma 4.3 that

$$|L(v)[L, R]| > \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n} - 1}{2} - 7000\epsilon^{1/4}\tilde{n}^2 > \left(\frac{2}{9} - 7010\epsilon^{1/4}\right)\tilde{n}^2. \tag{18}$$

It follows from Algorithm 2 that either $C_v \subset A^{i+1}$ or $C_v \subset B^{i+1}$, so by Claim 4.8

$$|L||R| \leq |A^{i+1}||B^{i+1}| - \alpha|A^{i+1}| > \left(\frac{1}{3} + 4\epsilon^{1/4}\right)\left(\frac{2}{3} - 4\epsilon^{1/4}\right)\tilde{n}^2 - \alpha\left(\frac{1}{3} - 4\epsilon^{1/4}\right)\tilde{n},$$

which together with (18) implies

$$\begin{aligned} \left(\frac{2}{9} - 7010\epsilon^{1/4}\right)\tilde{n}^2 &< |L(v)[L, R]| < |L||R| \\ &< \left(\frac{1}{3} + 4\epsilon^{1/4}\right)\left(\frac{2}{3} - 4\epsilon^{1/4}\right)\tilde{n}^2 - \alpha\left(\frac{1}{3} - 4\epsilon^{1/4}\right)\tilde{n}. \end{aligned}$$

Consequently, $\alpha < 30000\epsilon^{1/4}\tilde{n}$. ∎

Our next claim shows that $L(v)$ is indeed bipartite.

Claim 4.14. $L(v)[R] = \emptyset$.

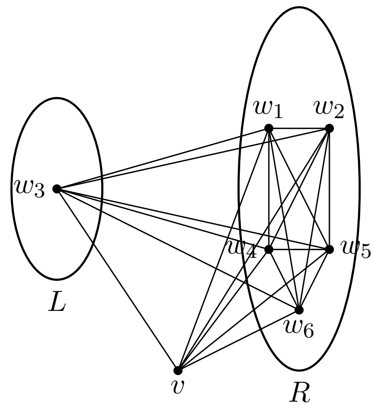


Figure 9: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_6\}$ is a copy of a 3-graph in M_2 since both vw_1w_2 and $w_3w_4w_5$ are in $\widetilde{\mathcal{H}}_i$ and they are disjoint.

Proof of Claim 4.14. Suppose there exists $w_1w_2 \in L(v)[R]$. It follows from $\widetilde{\mathcal{H}}_{i+1}[W'] = \widetilde{\mathcal{H}}_i[W']$ that $L(w_j)[W'] = L_{\widetilde{\mathcal{H}}_{i+1}}(w_j)[W']$ for $j \in \{1, 2\}$. By the induction hypothesis $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite, so $L(w_j)[W']$ is bipartite with parts L and R for $j \in \{1, 2\}$. It follows from Lemma 4.3, Claims 4.12, and 4.13 that

$$\begin{aligned} (L(v) \cap L(w_1) \cap L(w_2))\,[L, R] &> \frac{2}{9}\tilde{n}^2 - 3\left(10\epsilon^{1/2} + 7000\epsilon^{1/4} + 30000\epsilon^{1/4}\right)\tilde{n}^2 \\ &> \left(\frac{2}{9} - 120000\epsilon^{1/4}\right)\tilde{n}^2. \end{aligned} \tag{19}$$

Let $N_1 = N_{\widetilde{\mathcal{H}}_i}(v) \cap N_{\widetilde{\mathcal{H}}_i}(w_1) \cap N_{\widetilde{\mathcal{H}}_i}(w_2)$ and note that $N_1 \subset L \cup R$. Then (19) implies that $|N_1| > (1 - 500000\epsilon^{1/4})\tilde{n}$. Therefore, by Claim 4.8, $|N_1 \cap L| > (1/3 - 500004\epsilon^{1/4})\tilde{n}$, and hence there exists $w_3 \in N_1 \cap L$.

Let $N_2 = N_1 \cap N_{\widetilde{\mathcal{H}}_i}(w_3) \cap R$. A similar argument as above yields that $|N_2| > (2/3 - 600000\epsilon^{1/4})\tilde{n}$. Letting $R' = N_2$ in Claim 4.10 we obtain $d\left(L(w_3)[N_2]\right) \geq 1 - 400\epsilon^{1/4}$. Therefore, by Turán's theorem, $K_3 \subset L(w_3)[N_2]$ and we may assume that the vertex set of this K_3 is $\{w_4, w_5, w_6\}$. However, the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_6\}$ is a copy of a 3-graph in M_2, a contradiction. ∎

Now we are ready to finish the proof of Lemma 4.5.

By $(*)$, $N_{\widetilde{\mathcal{H}}_i}(v) \cap L \neq \emptyset$, so it follows from Claims 4.11 and 4.14 that $L(v)$ is a bipartite graph with two parts L and R. Since $\widetilde{\mathcal{H}}_{i+1}$ is semibipartite and $\widetilde{\mathcal{H}}_i[W'] = \widetilde{\mathcal{H}}_{i+1}[W']$, $\widetilde{\mathcal{H}}_i[W']$ is also semibipartite. Let $A^i = A^{i+1} \setminus C_v$ and $B^i = B^{i+1} \cup C_v$. Then $\widetilde{\mathcal{H}}_i$ is semibipartite with parts A^i and B^i, and this completes the proof. ∎

4.3 $\mathcal{G}^2$-colorable

In this section we will prove the following statement.

Lemma 4.15. *Suppose that $\mathcal{H}_t$ is $\mathcal{G}_6^2$-colorable and each part in $\mathcal{H}_t$ is a union of some equivalence classes in $\mathcal{H}_t$. Then $\widetilde{\mathcal{H}}_i$ is $\mathcal{G}_6^2$-colorable and each part in $\widetilde{\mathcal{H}}_i$ is a union of some equivalence classes in $\widetilde{\mathcal{H}}_i$ for all $0 \leq i \leq t$. In particular, $\mathcal{H}[W] = \widetilde{\mathcal{H}}_0$ is $\mathcal{G}_6^2$-colorable.*

Proof. Similar to Lemma 4.5, the proof of Lemma 4.15 is also by backward induction on i, and the base case is $i = t$ as $\widetilde{\mathcal{H}}_t = \mathcal{H}_t$. Now suppose that $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable for some $0 \le i \le t-1$, and we want to show that $\widetilde{\mathcal{H}}_i$ is also $\mathcal{G}_6^2$-colorable.

Since $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable, let

$$\mathcal{P} = \{A_1^{i+1}, A_2^{i+1}, A_3^{i+1}, A_4^{i+1}, B_1^{i+1}, B_2^{i+1}\}.$$

be the set of six parts in $\widetilde{\mathcal{H}}_{i+1}$ such that there is no edge between $A_1^{i+1}A_2^{i+1}B_1^{i+1}$, $A_1^{i+1}A_2^{i+1}B_2^{i+1}$, $A_3^{i+1}A_4^{i+1}B_1^{i+1}$, and $A_3^{i+1}A_4^{i+1}B_2^{i+1}$.

First we give a lower bound and an upper bound for $|S|$ for $S \in \mathcal{P}$. Let

$$y = (y_1, y_2, y_3, y_4, y_5, y_6) = \left(\frac{|B_1^{i+1}|}{\tilde{n}}, \frac{|A_1^{i+1}|}{\tilde{n}}, \frac{|A_2^{i+1}|}{\tilde{n}}, \frac{|B_2^{i+1}|}{\tilde{n}}, \frac{|A_3^{i+1}|}{\tilde{n}}, \frac{|A_4^{i+1}|}{\tilde{n}}\right),$$

Recall from Lemma 3.3 that

$$p_{\mathcal{G}_6^2}(x) = \sum_{ijk \subset [6]} x_i x_j x_k - (x_1x_2x_3 + x_2x_3x_4 + x_4x_5x_6 + x_5x_6x_1),$$

and it is easy to see that (similar to the proof of Lemma 2.1)

$$|\widetilde{\mathcal{H}}_{i+1}| \le p_{\mathcal{G}_6^2}(y)\tilde{n}^3. \tag{20}$$

Claim 4.16. *$|S - \tilde{n}/6| < \tilde{n}/100$ for all $S \in \mathcal{P}$.*

Proof of Claim 4.16. It suffice to show that $|y_i - 1/6| < 1/100$ for all $i \in [6]$. Let $\hat{x} = (1/6, \dots, 1/6) \in \Delta^5$, $B(\hat{x}, 1/100) = \left\{x \in \Delta^5 : |x - \hat{x}| < 1/100\right\}$, and $\Delta' = \Delta^5 \setminus B(\hat{x}, 1/100)$, where $|x - \hat{x}|$ is the distance between x and $\hat{x}$ in the Euclidean space. Since $B(\hat{x}, 1/100)$ is open in Δ^5, it follows that Δ' is closed in Δ^5. Therefore, there exists $x' \in \Delta'$ such that

$$p_{\mathcal{G}_6^2}(x') = \sup\left\{p_{\mathcal{G}_6^2}(x) : x \in \Delta'\right\}.$$

The proof of Lemma 3.3 implies that $\hat{x} \in \Delta^5$ is the unique point where $p_{\mathcal{G}_6^2}(x)$ attains its maximum on Δ^5, and it follows that $p_{\mathcal{G}_6^2}(x') < p_{\mathcal{G}_6^2}(\hat{x})$.

We may assume that $\epsilon > 0$ is sufficiently small that $10\epsilon^{1/2} < p_{\mathcal{G}_6^2}(\hat{x}) - p_{\mathcal{G}_6^2}(x')$. If $y_i \le 1/6 - 1/100$ for some $i \in [6]$, then $y \in \Delta'$, and hence by (20),

$$\begin{aligned}|\widetilde{\mathcal{H}}_{i+1}| \le p_{\mathcal{G}_6^2}(y) \le p_{\mathcal{G}_6^2}(x')\tilde{n}^3 &< \left(p_{\mathcal{G}_6^2}(\hat{x}) - 10\epsilon^{1/2}\right)\tilde{n}^3 \\ &= \left(\frac{2}{27} - 10\epsilon^{1/2}\right)\tilde{n}^3 < \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}}{3},\end{aligned}$$

which contradicts Lemma 4.3. ∎

We now use Claim 4.16 to prove much better bounds on $|S|$ for $S \in \mathcal{P}$.

Claim 4.17. *$||S| - \tilde{n}/6| < 10\epsilon^{1/4}\tilde{n}$ for all $S \in \mathcal{P}$.*

Proof of Claim 4.17. Let $\gamma = \max\{|y_i - 1/6| : i \in [6]\}$ and without loss of generality we may assume $|y_1 - 1/6| = \gamma$. It suffices to show that $\gamma < 10\epsilon^{1/4}$ and notice that Claim 4.16 already shows that $\gamma < 1/100$. Let $p(x) := p(x_1, x_2, x_3, x_4, x_5)$ be the function obtained from $p_{\mathcal{G}_6^2}(x)$ by substituting $x_6 = 1 - \sum_{i\in[5]} x_i$ into it. Consider the Taylor expansion of $p(x)$ at $x_0 = (1/6, 1/6, 1/6, 1/6, 1/6)$, which is

$$\begin{aligned} p(x) &= p(x_0) + \nabla p(x_0) \cdot (x - x_0)^T + \frac{1}{2}(x - x_0) \cdot H(x_0) \cdot (x - x_0)^T \\ &\quad + \sum_{i,j,k\in[5]} c_{ijk} \frac{\partial^3 p(x_0)}{\partial x_i \partial x_j \partial x_k} \left(x_i - \frac{1}{6}\right)\left(x_j - \frac{1}{6}\right)\left(x_k - \frac{1}{6}\right) \\ &\le \frac{2}{27} + \frac{1}{2}(x - x_0)H(x_0)(x - x_0)^T + |x - x_0|^3 \end{aligned}$$

where $H(x_0) = \left(\partial^2 p(x_0)/\partial x_i \partial x_j\right)_{i,j\in[5]}$ is the Hessian matrix of $p(x)$ at x_0, and

$$c_{ijk} = \begin{cases} 1/6 & i = j = k \\ 1 & i \ne j, j \ne k, k \ne i \\ 1/2 & \text{otherwise} \end{cases}$$

and we used the fact that $\nabla p(x_0) = (0, \dots, 0)$. Since the eigenvalues of

$$H(x_0) = \begin{pmatrix} -1 & -2/3 & -2/3 & -1/3 & -1/3 \\ -2/3 & -4/3 & -1 & -2/3 & -1/3 \\ -2/3 & -1 & -4/3 & -2/3 & -1/3 \\ -1/3 & -2/3 & -2/3 & -1 & -1/3 \\ -1/3 & -1/3 & -1/3 & -1/3 & -2/3 \end{pmatrix}$$

are $\left\{-\sqrt{2} - 2, -2/3, \sqrt{2} - 2, -1/3, -1/3\right\}$, $H(x_0)$ is negative-definite. Therefore, by the Min-max principle,

$$(x - x_0) \cdot H(x_0) \cdot (x - x_0)^T \le -\frac{1}{3}|x - x_0|^2,$$

and it follows that

$$p(x) \le \frac{2}{27} - \frac{1}{6}|x - x_0|^2 + |x - x_0|^3. \tag{21}$$

Since (21) is true for all $x \in \mathbb{R}^5$ and, in particular, for $\tilde{y} = (y_1, \dots, y_5)$,

$$\begin{aligned} p(\tilde{y}) \le \frac{2}{27} - \frac{1}{6}|\tilde{y} - x_0|^2 + |\tilde{y} - x_0|^3 &\le \frac{2}{27} - \frac{1}{6}|\tilde{y} - x_0|^2 + \sqrt{5}\gamma|\tilde{y} - x_0|^2 \\ &< \frac{2}{27} - \frac{1}{6}|\tilde{y} - x_0|^2 + \frac{\sqrt{5}}{100}|\tilde{y} - x_0|^2 \\ &< \frac{2}{27} - \frac{1}{10}|\tilde{y} - x_0|^2, \end{aligned} \tag{22}$$

and here we used $|\tilde{y} - x_0| \le \sqrt{5\gamma^2} = \sqrt{5}\gamma$ and $\gamma < 1/100$. Lemma 4.3 and (20) and (22) imply

$$\left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}}{3} < |\widetilde{H}_{i+1}| < \left(\frac{2}{27} - \frac{1}{10}|\tilde{y} - x_0|^2\right)\tilde{n}^3 < \left(\frac{2}{27} - \frac{1}{10}\gamma^2\right)\tilde{n}^3,$$

which implies that $\gamma < 10\epsilon^{1/4}$. ∎

Lemma 4.1 implies that either $\widetilde{\mathcal{H}}_i = \widetilde{\mathcal{H}}_{i+1}$ or there exists two equivalence classes V and U in $\widetilde{\mathcal{H}}_i$ such that $\widetilde{\mathcal{H}}_i$ is obtained from $\widetilde{\mathcal{H}}_{i+1}$ by symmetrizing V to U. In the former case, there is nothing to prove, so we may assume that we are in the later case. Suppose

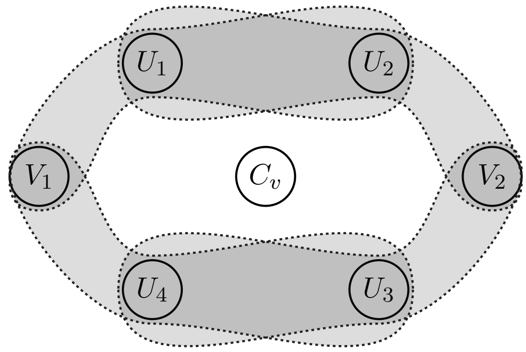


Figure 10: $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing C_v to some equivalence class C_u that is contained in some set in $\mathcal{P}'$. Dashed lines indicate that there is no edge between these parts in $\widetilde{\mathcal{H}}_i$.

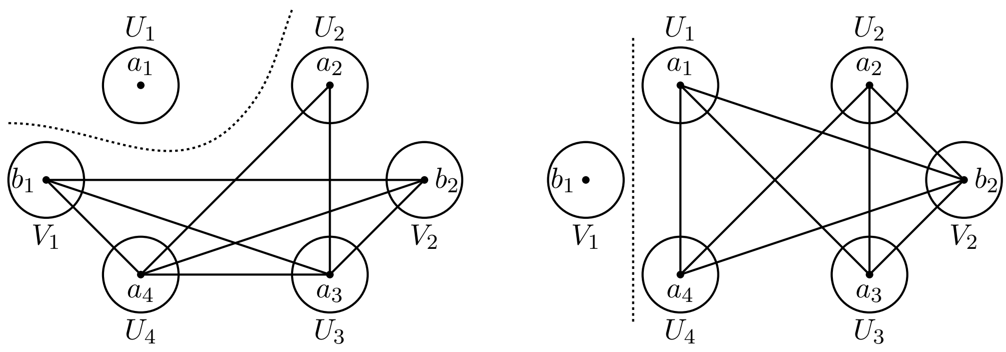
that V is the equivalence class of $v \in W$ and U is the equivalence class of $u \in W$ (in $\widetilde{\mathcal{H}}_i$), i.e. $V = C_v$ and $U = C_u$.

Notice that $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing C_v to C_u where C_u is contained in some $S \in \mathcal{P}$. Therefore, $C_v \subset S$ for some $S \in \mathcal{P}$, and in particular, Claim 4.17 implies that $|C_v| \leq (1/6 + 10\epsilon^{1/4})\tilde{n}$. In the rest of the proof we will focus on the structure of $\widetilde{\mathcal{H}}_i$. Let $U_j = A_j^{i+1} \setminus C_v$ for $1 \leq j \leq 4$, $V_j = B_j^{i+1} \setminus C_v$ for $1 \leq j \leq 2$, and $W' = W \setminus C_v$. Let

$$\mathcal{P}' = \{U_1, U_2, U_3, U_4, V_1, V_2\}.$$

Notice that no set in $\mathcal{P}'$ is the empty set, and in $\mathcal{P}'$ all but at most one set U_j or V_j is the same as A_j^{i+1} or B_j^{i+1}, respectively.

First we will prove several claims about U_1 and V_1, since U_1 is a representative for sets in $\{U_1, \ldots, U_4\}$ and V_1 is a representative for sets in $\{V_1, V_2\}$. Actually, similar claims hold for all sets in $\mathcal{P}'$ and we will omit the proof of them, since the proof is the same as the proof of either U_1 or V_1.



(a) The graph G_U, and $\widetilde{G}_U$ a blow up of G_U by replacing each vertex in $V(G_U)$ with the set in $\mathcal{P}'$ that contains it.

(b) The graph G_V, and $\widetilde{G}_V$ a blow up of G_V by replacing each vertex in $V(G_V)$ with the set in $\mathcal{P}'$ that contains it.

Figure 11: Graphs G_U and G_V.

Choose $a_j \in U_j$ for $1 \leq j \leq 4$ and $b_j \in V_j$ for $1 \leq j \leq 2$. Let G_U be a graph on $\{a_2, a_3, a_4, b_1, b_2\}$ with edge set

$$\{a_2a_3, a_2a_4, a_3a_4, a_3b_1, a_3b_2, a_4b_1, a_4b_2, b_1b_2\},$$

and let $\widetilde{G}_U$ be a blow up of G_U with a_j replaced by U_j for $j \in \{2, 3, 4\}$ and b_j replaced by V_j for $j \in \{1, 2\}$, and edges in G_U are replaced by the corresponding complete bipartite graph. Let G_V be a graph on $\{a_1, a_2, a_3, a_4, b_2\}$ with edge set

$$\{a_1a_3, a_1a_4, a_1b_2, a_2a_3, a_2a_4, a_2b_2, a_3b_2, a_4b_2\},$$

and let $\widetilde{G}_V$ be a blow up of G_V with a_j replaced by U_j for $j \in \{1, 2, 3, 4\}$ and b_2 replaced by V_2, and edges in G_V are replaced by the corresponding complete bipartite graph.

For every $w \in W$, let $L(w) = L_{\widetilde{\mathcal{H}}_i}(w)$ and $N(w) = N_{\widetilde{\mathcal{H}}_i}(w)$. Since $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable and $\widetilde{\mathcal{H}}_i[W'] = \widetilde{\mathcal{H}}_{i+1}[W']$, it follows that $L(w)[W'] \subset \widetilde{G}_U$ for all $w \in U_1$, and similarly, $L(w')[W'] \subset \widetilde{G}_V$ for all $w' \in V_1$. For every $w \in U_1$ and $w' \in V_1$, let

$$M_U(w) = \left\{ w_1w_2 \in \widetilde{G}_U : w_1w_2 \not\in L(w)[W'] \right\},$$

and

$$M_V(w') = \left\{ w_1w_2 \in \widetilde{G}_V : w_1w_2 \not\in L(w')[W'] \right\}.$$

Sets in $M_U(w)$ are called missing edges of $L(w)[W']$ and sets in $M_V(w')$ are called missing edges of $L(w')[W']$.

The following claim is an immediate consequence of Lemma 4.3.

Claim 4.18. *$|M_U(w)| \le 30\epsilon^{1/4}\tilde{n}^2$ for all $w \in U_1$, and $|M_V(w')| \le 30\epsilon^{1/4}\tilde{n}^2$ for all $w' \in V_1$.*

Proof of Claim 4.18. It suffice to prove $|M_U(w)| \le 30\epsilon^{1/4}\tilde{n}^2$ for all $w \in U_1$ since the proof for the other part is similar.

Fix $w \in U_1$. Let $\widehat{G}_U$ be a blow up of G_U by replacing each vertex in $V(G_U)$ with the set in $\mathcal{P}$ that contains it. Since $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable, $L_{\widetilde{\mathcal{H}}_{i+1}}(w) \subset \widehat{G}_U$. On the other hand, since $L_{\widetilde{\mathcal{H}}_i}(w)[W'] = L_{\widetilde{\mathcal{H}}_{i+1}}(w)[W']$, it follows from $|G_U| = 8$, Lemma 4.3 and Claim 4.17 that

$$\begin{aligned}
|M_U(w)| = |\widetilde{G}_U \setminus L_{\widetilde{\mathcal{H}}_i}(w)[W']| &\le |\widehat{G}_U \setminus L_{\widetilde{\mathcal{H}}_{i+1}}(w)| \\
&= |\widehat{G}_U| - |L_{\widetilde{\mathcal{H}}_{i+1}}(w)| \\
&< 8\left(\frac{1}{6} + 10\epsilon^{1/4}\right)^2 \tilde{n}^2 - \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}-1}{2} \\
&< 30\epsilon^{1/4}\tilde{n}^2.
\end{aligned}$$

∎

Claim 4.19. *Let $P \in \{U_1, V_1\}$ and $w \in P$. Then $|N(w) \cap (W' \setminus P)| > |W' \setminus P| - 400\epsilon^{1/4}\tilde{n}$.*

Proof. Without loss of generality, we may assume that $P = U_1$, since the proof for $P = V_1$ is similar. Let $w \in P$ and $W'' = W' \setminus U_1$. Since C_v is contained in exactly one set in $\mathcal{P}$, it follows from Claim 4.8 that all but at most one set in $\mathcal{P}'$ have size at least $\left(1/6 - 10\epsilon^{1/4}\right)\tilde{n}$. On the other hand, since $\delta(G_U) \ge 2$ and $\widetilde{G}_U$ is a blow up of G_U, we obtain

$$\delta(\widetilde{G}_U) > \left(\frac{1}{6} - 10\epsilon^{1/4}\right)\tilde{n}.$$

By Claim 4.18, the number of vertices in W'' with degree 0 in $L(w)[W']$ is at most

$$\frac{2|M_U(w)|}{\delta(\widetilde{G}_U)} < \frac{60\epsilon^{1/4}\tilde{n}^2}{(1/6 - 10\epsilon^{1/4})\tilde{n}} < 400\epsilon^{1/4}\tilde{n}.$$

∎

Recall that $\widetilde{\mathcal{H}}_{i+1}$ is obtained from $\widetilde{\mathcal{H}}_i$ by symmetrizing C_v to C_u, where C_v and C_u are equivalence classes of v and u in $\widetilde{\mathcal{H}}_i$, respectively. Let P_u denote the set in $\mathcal{P}'$ that contains u and notice that $P_u \cup C_v$ is a set in $\mathcal{P}$. So Claim 4.8 implies that $|P_u \cup C_v| \le (1/6+10\epsilon^{1/4})\tilde{n}$.

Claim 4.20. *Suppose that $|C_v| > \tilde{n}/12$. Then every vertex in $W' \setminus P_u$ is adjacent to all vertices in C_v in $\widetilde{\mathcal{H}}_i$.*

Proof. Without loss of generality we may assume that $P_u = U_2$, since the proof for the other cases is similar. Since $|P_u \cup C_v| \le (1/6 + 10\epsilon^{1/4})\tilde{n}$, it follows from our assumption that $|P_u| < (1/6 + 10\epsilon^{1/4})\tilde{n} - \tilde{n}/12 < \tilde{n}/10$. Let $w \in W' \setminus P_u$, and suppose that w is not adjacent to any vertex in C_v. Without loss of generality we may assume that $w \in U_1$, since the proof for the other cases is similar.

Since $\widetilde{\mathcal{H}}_i[W'] = \widetilde{\mathcal{H}}_{i+1}[W']$ and $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable, $L_{\widetilde{\mathcal{H}}_i}(w)[W'] \subset \widetilde{G}_U$. On the other hand, since $N_{\widetilde{\mathcal{H}}_i}(w) \cap C_v = \emptyset$, we actually have $L_{\widetilde{\mathcal{H}}_i}(w) \subset \widetilde{G}_U$. It follows from the definition of $\widetilde{G}_U$ and Claim 4.8 and $|U_2| = |P_u| < \tilde{n}/10$ that

$$|L_{\widetilde{\mathcal{H}}_i}(w)| \le |\widetilde{G}_U| < 6\left(\frac{1}{6} + 10\epsilon^{\frac{1}{4}}\right)^2 \tilde{n}^2 + 2 \times \frac{\tilde{n}}{10}\left(\frac{1}{6} + 10\epsilon^{\frac{1}{4}}\right)\tilde{n} < \left(\frac{2}{9} - 10\epsilon^{1/2}\right)\tilde{n}^2,$$

which contradicts Lemma 4.3.

Therefore, w is adjacent to some vertex in C_v in $\widetilde{\mathcal{H}}_i$. Since C_v is an equivalence class in $\widetilde{\mathcal{H}}_i$, w is adjacent to all vertices in C_v in $\widetilde{\mathcal{H}}_i$. ∎

Recall that $v \in C_v$ by the definition of C_v.

Claim 4.21. *$L(v)[S] = \emptyset$ for all $S \in \mathcal{P}'$.*

Proof. Suppose that $L(v)[S] \neq \emptyset$ for some $S \in \mathcal{P}'$ and $w_1w_2 \in L(v)[S]$. Without loss of generality we may assume that $S = U_1$, since the proof for the other cases is similar. It follows from Claim 4.19 that

$$|N(w_1) \cap N(w_2) \cap (W' \setminus U_1)| > |W' \setminus U_1| - 800\epsilon^{1/4}\tilde{n}. \tag{23}$$

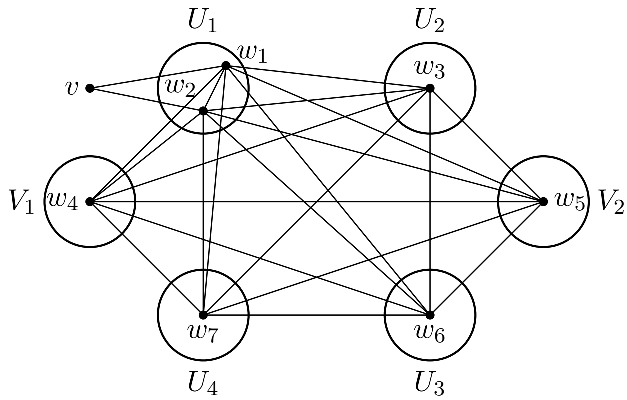


Figure 12: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{w_1, \ldots, w_7\}$ is a copy of a 3-graph in M_2, since $\{w_1w_5w_6, w_1w_6w_7, w_2w_5w_7\} \subset \widetilde{\mathcal{H}}_i$ and $\tau(\{w_1w_5w_6, w_1w_6w_7, w_2w_5w_7\}) > 1$.

Suppose that $|W' \setminus U_1| > 11\tilde{n}/15$. Then by Claim 4.17, every set in $\{U_2, U_3, U_4, V_1, V_2\}$ is of size at least $\tilde{n}/20$. Equation (23) implies that $|N(w_1) \cap N(w_2) \cap U_2| > |U_2| - 400\epsilon^{1/4}\tilde{n}$, so there exists $w_3 \in N(w_1) \cap N(w_2) \cap U_2$. Similarly, there exists $w_4 \in N(w_1) \cap N(w_2) \cap N(w_3) \cap V_1$.

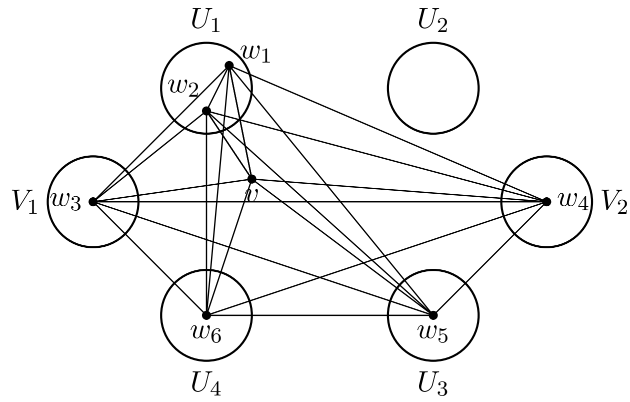


Figure 13: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{w_1, \ldots, w_7\}$ is a copy of a 3-graph in M_2, since $\{vw_1w_2, w_1w_5w_6, w_2w_5w_6\} \subset \widetilde{\mathcal{H}}_i$ and $\tau(\{vw_1w_2, w_1w_5w_6, w_2w_5w_6\}) > 1$.

Let $V_2' = N(w_1) \cap N(w_2) \cap N(w_3) \cap N(w_4) \cap V_2$, $U_3' = N(w_1) \cap N(w_2) \cap N(w_3) \cap N(w_4) \cap U_3$, and $U_4' = N(w_1) \cap N(w_2) \cap N(w_3) \cap N(w_4) \cap U_4$. Claim 4.19 implies that $|V_2'| > |V_2| - 1600\epsilon^{1/4}\tilde{n} > \tilde{n}/20 - 1600\epsilon^{1/4}\tilde{n}$, $|U_3'| > \tilde{n}/20 - 1600\epsilon^{1/4}\tilde{n}$, and $|U_4'| > \tilde{n}/20 - 1600\epsilon^{1/4}\tilde{n}$. Moreover, Claim 4.18 implies that the number of missing edges of $L(w_1)$ (resp. $L(w_2)$) that are contained in $V_2' \cup U_3' \cup U_4'$ is less than $30\epsilon^{1/4}\tilde{n}^2$. Therefore, there exists $w_5 \in V_2'$, $w_6 \in U_3'$, and $w_7 \in U_4'$ such that $\{w_5w_6, w_5w_7, w_6w_7\} \subset L(w_1) \cap L(w_2)$. However, this implies that $\widetilde{\mathcal{H}}_i$ contains a copy of a 3-graph in M_2 with core $\{w_1, \ldots, w_7\}$, a contradiction.

Suppose that $|W' \setminus U_1| \le 11\tilde{n}/15$. Then $|C_v| \ge \tilde{n} - (1/6 + 10\epsilon^{1/4})\tilde{n} - 11\tilde{n}/15 > \tilde{n}/12$ and $P_u \ne U_1$. Without loss of generality we may assume that $P_u = U_2$, since the proof for the other case is similar. By Claim 4.20, every vertex in $U_3 \cup U_4 \cup V_1 \cup V_2$ is adjacent to v. Similar to the argument above, there exists $w_3 \in N(w_1) \cap N(w_2) \cap N(v) \cap V_1$ and $w_4 \in N(w_1) \cap N(w_2) \cap N(w_3) \cap N(v) \cap V_2$. Let $U_3' = N(w_1) \cap N(w_2) \cap N(w_3) \cap N(w_4) \cap N(v) \cap U_3$, and $U_4' = N(w_1) \cap N(w_2) \cap N(w_3) \cap N(w_4) \cap N(v) \cap U_4$. Then Claim 4.9 implies that

$$|U_3'| > |U_3| - 4 \times 400\epsilon^{1/4}\tilde{n} \overset{\text{Claim 4.17}}{>} \left(\frac{1}{6} - 10\epsilon^{1/4}\right)\tilde{n} - 1600\epsilon^{1/4}\tilde{n} > \left(\frac{1}{6} - 1610\epsilon^{1/4}\right)\tilde{n},$$

and similarly, $|U_4'| > \tilde{n}/6 - 1610\epsilon^{1/4}\tilde{n}$. Claim 4.18 implies that the number of missing edges of $L(w_1)$ (resp. $L(w_2)$) that are contained in $U_3' \cup U_4'$ is less than $30\epsilon^{1/4}\tilde{n}^2$. Therefore, there exists $w_5 \in U_3'$ and $w_6 \in U_4'$ such that $w_5w_6 \in L(w_1) \cap L(w_2)$. However, this implies that $\widetilde{\mathcal{H}}_i$ contains a copy of a 3-graph in M_2 with core $\{v, w_1, ..., w_6\}$, a contradiction. ∎

Claim 4.22. *There is at most one set $S \in \mathcal{P}'$ such that $|N(v) \cap S| < \tilde{n}/48$.*

Proof of Claim 4.22. Let $U_j' = N(v) \cap U_j$ for $1 \le j \le 4$ and $V_j' = N(v) \cap V_j$ for $1 \le j \le 2$. By Claim 4.21, $L(v)$ is a 6-partite graph (not necessarily complete) with the set of parts $\mathcal{P}'' := \{U_1', U_2', U_3', U_4', V_1', V_2'\}$. Suppose that there are at least two sets in $\mathcal{P}''$ have size at most $\tilde{n}/48$. Then, by Claim 4.8,

$$\begin{aligned}|L(v)| &\le 6\left(\frac{1}{6} + 10\epsilon^{1/4}\right)^2 \tilde{n}^2 + \left(\frac{\tilde{n}}{48}\right)^2 + 8 \times \frac{\tilde{n}}{48}\left(\frac{1}{6} + 10\epsilon^{1/4}\right)\tilde{n} \\ &< \left(\frac{2}{9} - 10\epsilon^{1/2}\right)\tilde{n}^2,\end{aligned}$$

which contradicts Lemma 4.3. ∎

Claim 4.23. *There exists a set $S \in \mathcal{P}'$ such that $N(v) \cap S = \emptyset$.*

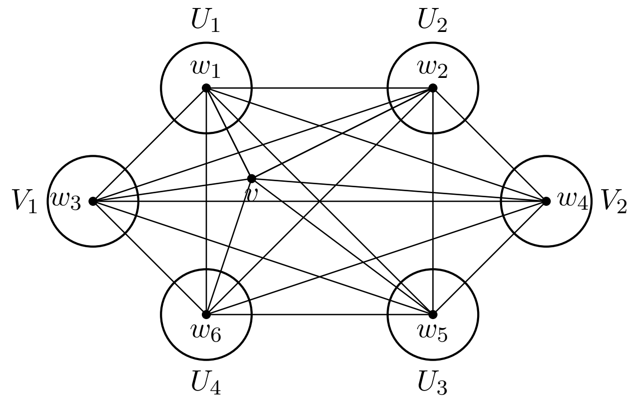


Figure 14: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{w_1, \ldots, w_7\}$ is a copy of a 3-graph in M_2, since $\{w_1w_4w_5, w_1w_5w_6, w_2w_4w_6\} \subset \widetilde{\mathcal{H}}_i$ and $\tau(\{w_1w_4w_5, w_1w_5w_6, w_2w_4w_6\}) > 1$.

Proof of Claim 4.23. By Claim 4.22, there are at least five sets $S \in \mathcal{P}'$ with $|S \cap N(v)| \geq \tilde{n}/48$. Without loss of generality we may assume that every set $S \in \{U_2, U_3, U_4, V_1, V_2\}$ satisfies $|S \cap N(v)| \geq \tilde{n}/48$, since the other case can be proved similarly. Our goal is to show that $N(v) \cap U_1 = \emptyset$.

Suppose that $N(v) \cap U_1 \neq \emptyset$ and let $w_1 \in N(v) \cap U_1$. Let $U_2' = N(v) \cap N(w_1) \cap U_2$, and $V_1' = N(v) \cap N(w_1) \cap V_1$. By Claim 4.19,

$$|U_2'| \geq |N(v) \cap U_2| - 400\epsilon^{1/4}\tilde{n} > \frac{\tilde{n}}{48} - 400\epsilon^{1/4}\tilde{n},$$

therefore, there exists $w_2 \in U_2'$. A similar argument as above implies that there exists $w_3 \in N(w_2) \cap V_1'$. Let $V_2' = N(v) \cap V_2 \cap N(w_1) \cap N(w_2) \cap N(w_3)$, $U_3' = N(v) \cap U_3 \cap N(w_1) \cap N(w_2) \cap N(w_3)$, and $U_4' = N(v) \cap U_4 \cap N(w_1) \cap N(w_2) \cap N(w_3)$. Claim 4.19 implies that

$$|V_2'| > |N(v) \cap V_2| - 3 \times 400\epsilon^{1/4}\tilde{n} > \frac{\tilde{n}}{48} - 1200\epsilon^{1/4}\tilde{n},$$

$|U_3'| > \tilde{n}/48 - 1200\epsilon^{1/4}\tilde{n}$, and $|U_4'| > \tilde{n}/48 - 1200\epsilon^{1/4}\tilde{n}$. Claim 4.18 implies that the number of missing edges of $L(w_1)$ (resp. $L(w_2)$) that are contained in $V_2' \cup U_3' \cup U_4'$ is at most $30\epsilon^{1/4}\tilde{n}^2$. Therefore, there exists $w_4 \in V_2'$, $w_5 \in U_3'$, and $w_6 \in U_4'$ such that $\{w_4w_5, w_4w_6, w_5w_6\} \subset L(w_1) \cap L(w_2)$. However, this implies that $\widetilde{\mathcal{H}}_i$ contains a copy of some 3-graph in M_2 with core $\{v, w_1, \ldots, w_6\}$, a contradiction. ∎

Claim 4.24. *There are five sets $S \in \mathcal{P}$ satisfying $|N(v) \cap S| \geq \tilde{n}/13$.*

Proof of Claim 4.24. Let $U_j' = N(v) \cap U_j$ for $1 \leq j \leq 4$ and $V_j' = N(v) \cap V_j$ for $1 \leq j \leq 2$. By Claim 4.21, $L(v)$ is a 6-partite graph (not necessarily complete) with the set of parts $\mathcal{P}'' := \{U_1', U_2', U_3', U_4', V_1', V_2'\}$. Claim 4.23 implies that there exists $S' \in \mathcal{P}''$ such that $S' = \emptyset$, so $L(v)$ is actually a 5-partite (not necessarily complete) graph, and our goal is to show that every set in $\mathcal{P}'' \setminus S'$ has size at least $\tilde{n}/13$.

Suppose for contradiction that there is a set in $\mathcal{P}'' \setminus S'$ which has size less than $\tilde{n}/13$. Then by Claim 4.17,

$$\begin{aligned}|L(v)| &\leq 6\left(\frac{1}{6} + 10\epsilon^{1/4}\right)^2 \tilde{n}^2 + 4 \times \frac{\tilde{n}}{13}\left(\frac{1}{6} + 10\epsilon^{1/4}\right)\tilde{n} \\ &< \left(\frac{2}{9} - 10\epsilon^{1/2}\right)\tilde{n}^2,\end{aligned}$$

which contradicts Lemma 4.3. ∎

Now we are ready to finish the proof of Lemma 4.15. By Claim 4.23, there exists $S \in \mathcal{P}'$ such that $S \cap N(v) = \emptyset$. Our goal is to show that $\widetilde{\mathcal{H}}_i$ is $\mathcal{G}_6^2$-colorable with the sets of parts $\widetilde{\mathcal{P}}$, where $\widetilde{\mathcal{P}}$ is obtained from $\mathcal{P}'$ by replacing S with $S \cup C_v$. Without loss of generality we may just consider two cases: $S = U_1$ and $S = V_1$.

Case 1: $N(v) \cap U_1 = \emptyset$.
Since $\widetilde{\mathcal{H}}_{i+1}[W'] = \widetilde{\mathcal{H}}_i[W']$ and $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable, $\widetilde{\mathcal{H}}_i[W']$ is also $\mathcal{G}_6^2$-colorable. If $L(v) \subset \widetilde{G}_U$, then let $A_1^i = A_1^{i+1} \cup C_v$, $A_j^i = A_j^{i+1} \setminus C_v$ for $j \in \{2,3,4\}$, and $B_j^i = B_j^{i+1} \setminus C_v$ for $j \in \{1,2\}$. It is easy to see that $\widetilde{\mathcal{H}}_i$ is $\mathcal{G}_6^2$-colorable with parts $\{A_1^i, A_2^i, A_3^i, A_4^i, B_1^i, B_2^i\}$, and we are done. So, we may assume that $L(v) \setminus \widetilde{G}_U \neq \emptyset$.

Let
$$B_v = \left\{ ww' \in L(v) : ww' \notin \widetilde{G}_U \right\},$$
and
$$M_v = \left\{ ww' \in \widetilde{G}_U : ww' \notin L(v) \right\}.$$
Sets in B_v are called bad edges of $L(v)$ and sets in M_v are called missing edges of $L(v)$. Since $B_v \neq \emptyset$, there exists $w_1w_2 \in B_v$. Claim 4.21 implies that either $w_1w_2 \in V_1 \times U_2$ or $w_1w_2 \in V_2 \times U_2$, and without loss of generality we may assume that $w_1 \in V_1$ and $w_2 \in U_2$.

Case 1.1: $|B_v| \leq \tilde{n}^2/10000$.

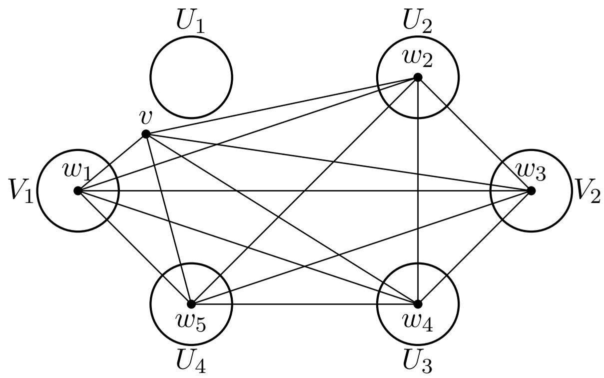


Figure 15: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_5\}$ is a copy of a 3-graph in M_3, since $\{vw_1w_2, vw_3w_4, vw_3w_5, vw_4w_5, w_1w_3w_4, w_1w_3w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset \widetilde{\mathcal{H}}_i$.

Then, by the fact that $|G_U| = 8$ and Lemma 4.3,
$$\begin{aligned} |M_v| = |\widetilde{G}_U \setminus L(v)| &= |\widetilde{G}_U| - |\widetilde{G}_U \cap L(v)| \\ &= |\widetilde{G}_U| - (|L(v)| - |B_v|) \\ &< 8\left(\frac{1}{6} + 10\epsilon^{1/4}\right)^2 \tilde{n}^2 - \left(\left(\frac{2}{9} - 10\epsilon^{1/2}\right)\tilde{n}^2 - |B_v|\right) \\ &< \frac{\tilde{n}^2}{10000} + 30\epsilon^{1/4}\tilde{n}^2. \end{aligned} \tag{24}$$
Let $L(vw_2) = L(v) \cap L(w_2)$ and $L(vw_1w_2) = L(v) \cap L(w_1) \cap L(w_2)$. Then (24), Lemma

4.3, and Claim 4.17 imply that

$$
\begin{aligned}
|L(vw_1w_2)[V_2 \cup U_3]| &\geq \left(\frac{2}{9} - 10\epsilon^{1/2}\right)\tilde{n}^2 - (|G_U|-1)\left(\frac{1}{6} + 10\epsilon^{\frac{1}{4}}\right)^2 \tilde{n}^2 \\
&\quad - \left(\frac{\tilde{n}^2}{10000} + 30\epsilon^{1/4}\tilde{n}^2\right) \\
&> \frac{\tilde{n}^2}{36} - \frac{\tilde{n}^2}{10000} - 50\epsilon^{1/4}\tilde{n}^2, \qquad (25)
\end{aligned}
$$

and similarly,

$$
|L(vw_1w_2)[V_2 \cup U_4]| > \frac{\tilde{n}^2}{36} - \frac{\tilde{n}^2}{10000} - 50\epsilon^{1/4}\tilde{n}^2, \qquad (26)
$$

and

$$
|L(vw_2)[U_3 \cup U_4]| > \frac{\tilde{n}^2}{36} - \frac{\tilde{n}^2}{10000} - 50\epsilon^{1/4}\tilde{n}^2. \qquad (27)
$$

Let

$$
S_1 = \left\{ w \in V_2 : |N_{L(vw_1w_2)}(w) \cap U_3| \geq \frac{\tilde{n}}{20} \text{ and } |N_{L(vw_1w_2)}(w) \cap U_4| \geq \frac{\tilde{n}}{20} \right\}.
$$

There exists $w_3 \in S_1$, since otherwise Claim 4.24 and Lemma 4.3 would imply

$$
\begin{aligned}
&|L(vw_1w_2)[V_2 \cup U_3]| + |L(vw_1w_2)[V_2 \cup U_4]| \\
&\leq |V_2|\left(|U_3| + |U_4|\right) - |V_2|\left(\min\{|U_3|, |U_4|\} - \frac{\tilde{n}}{20}\right) \\
&< 2\left(\frac{1}{6} + 10\epsilon^{\frac{1}{4}}\right)^2 \tilde{n}^2 - \frac{\tilde{n}}{13}\left(\frac{\tilde{n}}{13} - \frac{\tilde{n}}{20}\right) \\
&< \frac{\tilde{n}^2}{18} - \frac{\tilde{n}^2}{2000},
\end{aligned}
$$

which contradicts (25) and (26). Therefore, there exists $w_3 \in S_1$.

Let $U_3' = N_{L(vw_1w_2)}(w_3) \cap U_3$ and $U_4' = N_{L(vw_1w_2)}(w_3) \cap U_4$. By the definition of S_1, $|U_3'| \geq \tilde{n}/20$ and $|U_4'| \geq \tilde{n}/20$, and by (27), there exists $w_4 \in U_3'$ and $w_5 \in U_4'$ such that $w_4w_5 \in L(vw_2)$. Moreover, it follows from the definition of S_1 that $\{w_3w_4, w_3w_5\} \subset L(v) \cap L(w_1) \cap L(w_2)$.

Let F denote the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_5\}$. Then it follows from the argument above that

$$
\{vw_1w_2, vw_3w_4, vw_3w_5, vw_4w_5, w_1w_3w_4, w_1w_3w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset F.
$$

It is not hard to check that $F \not\subset \mathcal{G}_n^1$ and $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$ (using the following observations), so $F \in M_3$, a contradiction.

Since $\partial F \approx K_6$ and $\tau(F) > 1$, $F \not\subset \mathcal{G}_n^1$ for all $n \geq 6$. On the other hand, in order to show $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$, it suffices to show $F \not\subset \mathcal{G}_6^2$, which is equivalent to showing that $\overline{\mathcal{G}_6^2} \not\subset \overline{F}$. Notice that $\overline{\mathcal{G}_6^2}$ satisfies the following:

(1) For every $w \in \overline{\mathcal{G}_6^2}$ there exists $E \in \overline{\mathcal{G}_6^2}$ such that $w \notin E$.

(2) If $E \in \overline{\mathcal{G}_6^2}$, then $V(\overline{\mathcal{G}_6^2}) \setminus E \in \overline{\mathcal{G}_6^2}$ as well.

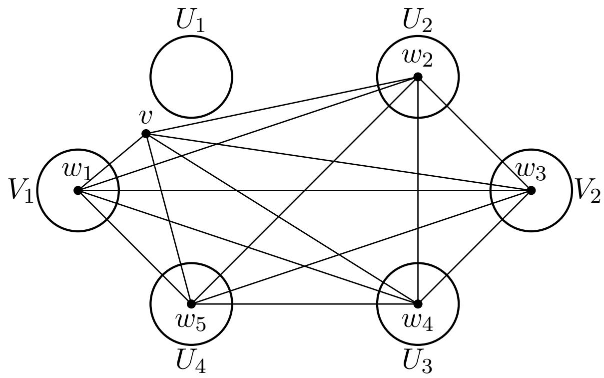


Figure 16: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_5\}$ is a copy of a 3-graph in M_3, since $\{vw_1w_2, w_1w_2w_3, w_1w_2w_4, w_1w_2w_5, w_1w_3w_4, w_1w_3w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset \widetilde{\mathcal{H}}_i$.

Case 1.2: $|B_v| > \tilde{n}^2/10000$.

Let
$$B = \left\{ww' \in B_v : w \in V_1 \text{ and } w' \in U_2\right\},$$
and without loss of generality we may assume that $|B| > |B_v|/2 > \tilde{n}^2/20000$. Let $x = |V_2 \cup U_3 \cup U_4|$ and note from Claim 4.24 that $\min\{|V_2|, |U_3|, |U_4|\} \geq \tilde{n}/13$. Define
$$S = \left\{ww' \in B : |N_{\widetilde{\mathcal{H}}_i}(ww') \cap (V_2 \cup U_3 \cup U_4)| < x - \epsilon^{1/4}\tilde{n}\right\}.$$
Since $\widetilde{\mathcal{H}}_{i+1}[W'] = \widetilde{\mathcal{H}}_i[W']$, if $ww'w'' \subset W'$ and $ww'w'' \notin \widetilde{\mathcal{H}}_i$, then $ww'w'' \notin \widetilde{\mathcal{H}}_{i+1}$. By Theorem 1.7, $|\widetilde{\mathcal{H}}_{i+1}| \leq 2\tilde{n}^3/27$, so it follows from Lemma 4.3 and Claim 4.8 that
$$|S|\epsilon^{1/4}\tilde{n} < \frac{2}{27}\tilde{n}^3 - \left(\frac{4}{9} - 10\epsilon^{1/2}\right)\binom{\tilde{n}}{3} < 10\epsilon^{1/2}\tilde{n}^3,$$
which implies that $|S| \leq 10\epsilon^{1/4}\tilde{n}^2$. Therefore, at least $\tilde{n}^2/30000$ edges in B have at least $x - \epsilon^{1/4}\tilde{n}$ neighbors in $V_2 \cup U_3 \cup U_4$.

Let $w_1w_2 \in B$ with $w_1 \in V_1, w_2 \in U_2$ such that w_1w_2 has at least $x - \epsilon^{1/4}\tilde{n}$ neighbors (in $\widetilde{\mathcal{H}}_i$) in $V_2 \cup U_3 \cup U_4$. Let $V_2' = N_{\widetilde{\mathcal{H}}_i}(w_1w_2) \cap V_2$, $U_3' = N_{\widetilde{\mathcal{H}}_i}(w_1w_2) \cap U_3$, and $U_4' = N_{\widetilde{\mathcal{H}}_i}(w_1w_2) \cap U_4$, and note that $|V_2'| > |V_2| - \epsilon^{1/4}\tilde{n} > \tilde{n}/13 - \epsilon^{1/4}\tilde{n}$, $|U_3'| > \tilde{n}/13 - \epsilon^{1/4}\tilde{n}$, and $|U_4'| > \tilde{n}/13 - \epsilon^{1/4}\tilde{n}$. Then by Claim 4.18, the number of missing edges of $L(w_1)$ (resp. $L(w_2)$) that are contained in $V_2' \cup U_3' \cup U_4'$ is at most $30\epsilon^{1/4}\tilde{n}^2$. Therefore, there exists $w_3 \in V_2'$, $w_4 \in U_3'$, and $w_5 \in U_4'$ such that $\{w_3w_4, w_3w_5, w_4w_5\} \subset L(w_2)$ and $\{w_3w_5, w_4w_5\} \subset L(w_1)$. Moreover, it follows from the definition of V_2', U_3', U_4' that $\{w_1w_2w_3, w_1w_2w_4, w_1w_2w_5\} \subset \widetilde{\mathcal{H}}_i$.

Let F denote the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_5\}$. Then it follows from the argument above that
$$\{vw_1w_2, w_1w_2w_3, w_1w_2w_4, w_1w_2w_5, w_1w_3w_4, w_1w_3w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset F.$$
It is not hard to check that $F \not\subset \mathcal{G}_n^1$ and $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$, a contradiction.

Case 2: $V_1 \cap N(v) = \emptyset$.
Since $\widetilde{\mathcal{H}}_{i+1}[W'] = \widetilde{\mathcal{H}}_i[W']$ and $\widetilde{\mathcal{H}}_{i+1}$ is $\mathcal{G}_6^2$-colorable, $\widetilde{\mathcal{H}}_i[W']$ is also $\mathcal{G}_6^2$-colorable. If $L(v) \subset \widetilde{G}_V$, then let $B_1^i = B_1^{i+1} \cup C_v$, $B_2^i = B_1^{i+1} \setminus C_v$, and $A_j^i = A_j^{i+1} \setminus C_v$ for $1 \leq j \leq 4$. It is easy to see that $\widetilde{\mathcal{H}}_i$ is $\mathcal{G}^2$-colorable with the set of parts $\{A_1^i, A_2^i, A_3^i, A_4^i, B_1^i, B_2^i\}$, and we are done. So we may assume that $L(v) \setminus \widetilde{G}_V \neq \emptyset$.

Let

$$\widetilde{B}_v = \left\{ ww' \in L(v) : ww' \notin \widetilde{G}_V \right\},$$

and

$$\widetilde{M}_v = \left\{ ww' \in \widetilde{G}_V : ww' \notin L(v) \right\}.$$

Sets in $\widetilde{B}_v$ are called bad edges of $L(v)$ and sets in $\widetilde{M}_v$ are called missing edges of $L(v)$. Since $\widetilde{B}_v \neq \emptyset$, there exists $w_1w_2 \in \widetilde{B}_v$. Claim 4.21 implies that either $w_1w_2 \in U_1 \times U_2$ or $w_1w_2 \in U_3 \times U_4$, and without loss of generality we may assume that $w_1 \in U_1$ and $w_2 \in U_2$.

Case 2.1: $|\widetilde{B}_v| \leq \tilde{n}^2/10000$.

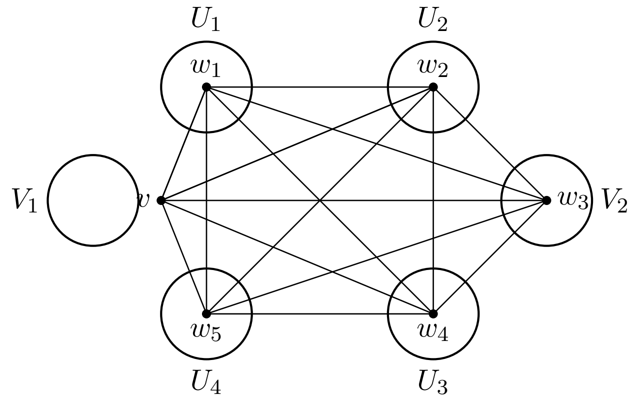


Figure 17: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \ldots, w_6\}$ is a copy of a 3-graph in M_3, since $\{vw_1w_2, vw_3w_4, vw_3w_5, w_1w_3w_4, w_1w_3w_5, w_1w_4w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset \widetilde{\mathcal{H}}_i$.

Similar to Case 1, there exists $w_3 \in V_2$, $w_4 \in U_3$, and $w_5 \in U_4$ such that $\{w_3w_4, w_3w_5\} \subset L(v)$ and $\{w_3w_4, w_3w_5, w_4w_5\} \subset L(w_1) \cap L(w_2)$.

Let F denote the induced subgraph of $\mathcal{H}_i[W]$ on $\{v, w_1, ..., w_5\}$. Then,

$$\{vw_1w_2, vw_3w_4, vw_3w_5, w_1w_3w_4, w_1w_3w_5, w_1w_4w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset F.$$

Therefore, $F \not\subset \mathcal{G}_n^1$ and $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$, and hence $F \in M_3$, a contradiction.

Case 2.2: $|\widetilde{B}_v| > \tilde{n}^2/10000$.

Similar to Case 1, there exists $w_3 \in V_2$, $w_4 \in U_3$, and $w_5 \in U_4$ such that $\{w_3, w_4, w_5\} \subset N_{\widetilde{\mathcal{H}}_i}(v) \cap N_{\widetilde{\mathcal{H}}_i}(w_1) \cap N_{\widetilde{\mathcal{H}}_i}(w_2)$, $\{w_4, w_5\} \subset N_{\widetilde{\mathcal{H}}_i}(w_1w_2)$, and $\{w_3w_4, w_3w_5, w_4w_5\} \subset L(w_1) \cap L(w_2)$.

Let F denote the induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, ..., w_5\}$. Then,

$$\{vw_1w_2, w_1w_2w_4, w_1w_2w_5, w_1w_3w_4, w_1w_3w_5, w_1w_4w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset F.$$

Therefore, $F \not\subset \mathcal{G}_n^1$ and $F \not\subset \mathcal{G}_n^2$ for all $n \geq 6$, and hence $F \in M_3$, a contradiction. ∎

5 Feasible Region of $\mathcal{M}$

In this section we consider the feasible region $\Omega(\mathcal{M})$ of $\mathcal{M}$ and prove Theorem 1.13. First, notice that our proof of Theorem 1.12 actually gives the following stronger statement.

Theorem 5.1. *For every sufficiently small $\epsilon > 0$ there exists n_0 such that the following holds for all $n \geq n_0$. Every $\mathcal{M}$-free 3-graph $\mathcal{H}$ with n vertices and at least $2n^3/27 - \epsilon n^3$ edges contains $W \subset V(\mathcal{H})$ with $|W| > n - 3\epsilon^{1/2}n$ such that $\delta(\mathcal{H}[W]) \geq 2n^2/9 - 20\epsilon^{1/2}n$ and $\mathcal{H}[W]$ is either semibipartite or $\mathcal{G}_6^2$-colorable.*

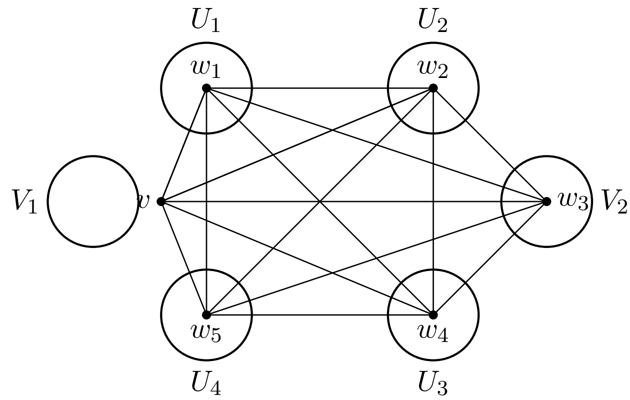


Figure 18: The induced subgraph of $\widetilde{\mathcal{H}}_i$ on $\{v, w_1, \dots, w_6\}$ is a copy of a 3-graph in M_3, since $\{vw_1w_2, w_1w_2w_4, w_1w_2w_5, w_1w_3w_4, w_1w_3w_5, w_1w_4w_5, w_2w_3w_4, w_2w_3w_5, w_2w_4w_5\} \subset \widetilde{\mathcal{H}}_i$.

Theorem 5.1 gives the following lemma.

Lemma 5.2. *Let $\epsilon > 0$ be sufficiently small and n (related to ϵ) be sufficiently large. Suppose that $\mathcal{H}$ is an $\mathcal{M}$-free 3-graph with n vertices and at least $2n^3/27 - \epsilon n^3$ edges. Then,*

$$\text{either } \left| |\partial\mathcal{H}| - \frac{5}{12}n^2 \right| < 100\epsilon^{1/4}n^2 \text{ or } \left| |\partial\mathcal{H}| - \frac{4}{9}n^2 \right| < 100\epsilon^{1/4}n^2.$$

Proof. Let $\epsilon > 0$ be sufficiently small and n (related to ϵ) be sufficiently large. Let $\mathcal{H}$ be an $\mathcal{M}$-free 3-graph with n vertices and at least $2n^3/27 - \epsilon n^3$ edges. By Theorem 5.1, there exists $W \subset V(\mathcal{H})$ with $|W| > n - 3\epsilon^{1/2}n$ such that $\delta(\mathcal{H}[W]) \geq 2n^2/9 - 20\epsilon^{1/2}n^2$ and $\mathcal{H}[W]$ is either semibipartite or $\mathcal{G}_6^2$-colorable. Let $Z = V(\mathcal{H}) \setminus W$, $\tilde{n} = |W|$, and $\widetilde{\mathcal{H}} = \mathcal{H}[W]$. Then,

$$|\widetilde{\mathcal{H}}| = \frac{1}{3}\sum_{w \in W} d_{\widetilde{\mathcal{H}}}(w) > \frac{1}{3}\left(n - 3\epsilon^{1/2}n\right)\left(\frac{2}{9}n^2 - 20\epsilon^{1/2}n^2\right) > \frac{2}{27}n^3 - 20\epsilon^{1/2}n^3. \tag{28}$$

Suppose that $\mathcal{H}[W]$ is semibipartite and let L and R denote the two parts of $\mathcal{H}[W]$ such that every $E \in \mathcal{H}[W]$ satisfies $|A \cap E| = 1$ and $|B \cap E| = 2$. Note from Claim 4.8 that

$$|L| = \frac{|W|}{3} \pm 4\epsilon^{1/4}|W| = \frac{n}{3} \pm 8\epsilon^{1/4}n \tag{29}$$

and

$$|L| = \frac{2|W|}{3} \pm 4\epsilon^{1/4}|W| = \frac{2n}{3} \pm 8\epsilon^{1/4}n. \tag{30}$$

First we prove the lower bound for $|\partial\mathcal{H}|$. Let

$$(\partial\widetilde{\mathcal{H}})[R] = \left\{uv \in \partial\widetilde{\mathcal{H}} : \{u, v\} \subset R\right\},$$

and

$$(\partial\widetilde{\mathcal{H}})[L, R] = \left\{uv \in \partial\widetilde{\mathcal{H}} : u \in L, v \in R\right\}.$$

Since $\widetilde{\mathcal{H}}$ is semibipartite,

$$|L||(\partial\widetilde{\mathcal{H}})[R]| \geq \sum_{v \in L} d_{\widetilde{H}}(v) = |\widetilde{H}|$$

and

$$|R||(\partial\widetilde{\mathcal{H}})[L,R]| \geq \sum_{u\in R} d_{\widetilde{H}}(u) = 2|\widetilde{H}|.$$

Consequently,

$$\begin{aligned}
|\partial\widetilde{\mathcal{H}}| = |(\partial\widetilde{\mathcal{H}})[R]| + |(\partial\widetilde{\mathcal{H}})[L,R]| &\geq \frac{|\widetilde{\mathcal{H}}|}{|L|} + \frac{2|\widetilde{\mathcal{H}}|}{|R|} \\
&\overset{(28),(29),(30)}{>} \frac{2n^3/27 - 20\epsilon^{1/2}n^3}{(1/3+8\epsilon^{1/4})n} + \frac{2(2n^3/27 - 20\epsilon^{1/2}n^3)}{(2/3+8\epsilon^{1/4})n} \\
&> \frac{4}{9}n^2 - 100\epsilon^{1/2}n^2.
\end{aligned}$$

Therefore, $|\partial\mathcal{H}| \geq |\partial\widetilde{\mathcal{H}}| > 4n^2/9 - 100\epsilon^{1/2}n^2$.

Next, we prove the upper bound for $|\partial\mathcal{H}|$. Let $v \in Z$ and suppose that $L_{\mathcal{H}}(v)[L] \neq \emptyset$. Then Claim 4.11 implies that $\mathcal{H}$ contains a 3-graph in M_2, a contradiction. Therefore, $L_{\mathcal{H}}(v)[L] = \emptyset$ for all $v \in Z$, and it follows that

$$\begin{aligned}
|\partial\mathcal{H}| &\leq \binom{|W|}{2} - \binom{|L|}{2} + |Z||W| + \binom{|Z|}{2} \\
&\leq \binom{n}{2} - \binom{|L|}{2} < \frac{1}{2}n^2 - \frac{1}{2}\left(\frac{1}{3} - 8\epsilon^{1/4}\right)n^2 < \frac{4}{9}n^2 + 100\epsilon^{1/4}n^2.
\end{aligned}$$

Therefore, if $\widetilde{\mathcal{H}}$ is semibipartite, then

$$\frac{4}{9}n^2 - 100\epsilon^{1/2}n^2 < |\partial\mathcal{H}| < \frac{4}{9}n^2 + 100\epsilon^{1/4}n^2.$$

Suppose that $\widetilde{\mathcal{H}}$ is $\mathcal{G}_6^2$-colorable and let $\mathcal{P} := \{A_1, A_2, A_3, A_4, B_1, B_2\}$ be the set of six parts of $\widetilde{\mathcal{H}}$ such that there is no edge between $A_1A_2B_1$, $A_1A_2B_2$, $A_3A_4B_1$, and $A_3A_4B_2$. Notice from Claim 4.17 that

$$|S| = \frac{|W|}{6} \pm 10\epsilon^{1/2}|W| = \frac{n}{6} \pm 10\epsilon^{1/4}n, \text{ for all } S \in \mathcal{P}. \tag{31}$$

First, we prove the lower bound for $|\partial\mathcal{H}|$. Since $\widetilde{\mathcal{H}}$ is $\mathcal{G}_6^2$-colorable, $\partial\widetilde{\mathcal{H}}$ is a 6-partite graph the set of six parts $\mathcal{P}$. Let $\mathcal{G}$ denote the complete 6-partite graph with the set of six parts $\mathcal{P}$ and let $\mathcal{G}$ denote the blow up of $\mathcal{G}_6^2$ with the set of six parts $\mathcal{P}$ such that there is no edge between $A_1A_2B_1$, $A_1A_2B_2$, $A_3A_4B_1$, and $A_3A_4B_2$. Notice that for every $e \in G \setminus \partial\widetilde{\mathcal{H}}$ there are at least $2(|W|/6 - 10\epsilon^{1/2}|W|)$ sets $E \in \mathcal{G} \setminus \widetilde{\mathcal{H}}$ such that $e \in E$. Therefore,

$$|G \setminus \partial\widetilde{\mathcal{H}}| \leq \frac{3|\mathcal{G} \setminus \widetilde{\mathcal{H}}|}{2(|W|/6 - 10\epsilon^{1/2}|W|)} \overset{(28),(31)}{<} \frac{3 \times 20\epsilon^{1/2}n^2}{2(n/6 - 10\epsilon^{1/4}n)} < 400\epsilon^{1/2}n^2,$$

and it follows that

$$|\partial\widetilde{\mathcal{H}}| > |G| - 400\epsilon^{1/2}n^2 \overset{(31)}{>} \binom{6}{2}\left(\frac{n}{6} - 10\epsilon^{1/4}n\right)^2 - 400\epsilon^{1/2}n^2 > \frac{5}{12}n^2 - 100\epsilon^{1/4}n^2$$

Therefore, $|\partial\mathcal{H}| \geq |\partial\widetilde{\mathcal{H}}| > 5n^2/12 - 100\epsilon^{1/2}n^2$.

Next, we prove the upper bound for $|\partial\mathcal{H}|$. Let $v \in Z$ and suppose that $L_{\mathcal{H}}(z)[S] \neq \emptyset$ for some $S \in \mathcal{P}$. Then Claim 4.21 implies that $\mathcal{H}$ contains a copy of a 3-graph in M_2, a contradiction. Therefore, $L_{\mathcal{H}}(z)[S] = \emptyset$ for all $S \in \mathcal{P}$, and it follows that

$$|\partial\mathcal{H}| \leq \frac{5}{12}|W|^2 + |Z||W| + \binom{|Z|}{2} < \frac{5}{12}n^2 + 100\epsilon^{1/4}n^2.$$

Therefore, if $\widetilde{\mathcal{H}}$ is $\mathcal{G}^2$-colorable, then

$$\frac{5}{12}n^2 - 100\epsilon^{1/2}n^2 < |\partial\mathcal{H}| < \frac{5}{12}n^2 + 100\epsilon^{1/4}n^2.$$

∎

Now we are ready to prove Theorem 1.13.

Proof of Theorem 1.13. Let

$$\mathcal{S}_n = \{A \subset [n] : 1 \in A\}.$$

Since $\mathcal{S}_n$ is $\mathcal{M}$-free and $|\partial\mathcal{S}_n| = \binom{n}{2}$, it follows from Observation 1.5 in [12] that $\mathrm{proj}\Omega(\mathcal{M}) = [0,1]$. On the other hand, it follows from Theorem 1.7 that $g(\mathcal{M}, x) \le 4/9$ for all $x \in [0,1]$ and $g(\mathcal{M}, 5/6) = g(\mathcal{M}, 8/9) = 4/9$.

Now suppose that $(\mathcal{H}_k)_{k=1}^{\infty}$ is a sequence of $\mathcal{M}$-free 3-graphs with $\lim_{k\to\infty} v(\mathcal{H}_k) = \infty$, $\lim_{k\to\infty} d(\partial\mathcal{H}_k) = x_0$, and $\lim_{k\to\infty} d(\mathcal{H}_k) = 4/9$. For any sufficiently small $\epsilon > 0$ and sufficiently n_0, there exists k_0 such that $v(\mathcal{H}_k) \ge n_0$ and $|\mathcal{H}_k| > 2(v(\mathcal{H}_k))^3/27 - \epsilon(v(\mathcal{H}_k))^3$ for all $k \ge k_0$. Therefore, by Lemma 5.2, for every $k \ge k_0$ either

$$\frac{8}{9} - 200\epsilon^{1/4} < \frac{|\partial\mathcal{H}_k|}{\binom{v(\mathcal{H}_k)}{2}} < \frac{8}{9} + 200\epsilon^{1/4}$$

or

$$\frac{5}{6} - 200\epsilon^{1/4} < \frac{|\partial\mathcal{H}_k|}{\binom{v(\mathcal{H}_k)}{2}} < \frac{5}{6} + 200\epsilon^{1/4}.$$

Letting $\epsilon \to 0$ we obtain either $x_0 = 8/9$ or $x_0 = 5/6$, and this completes the proof. ∎

6 Open Problems

Recall from Definitions 1.8 and 1.9 that the stability number $\xi(\mathcal{F})$ of a family $\mathcal{F}$ is the minimum integer t such that $\mathcal{F}$ is t-stable. We constructed a family $\mathcal{M}$ of 3-graphs and proved that $\xi(\mathcal{M}) = 2$. It seems natural to consider the following problems.

Problem 6.1. *For every pair (r,t) with $r \ge 3$ and $t \ge 2$, determine if there exists a family $\mathcal{F}$ of r-graphs with $\xi(\mathcal{F}) = t$.*

It follows from Kostochka's result [11] that $\xi(K_4^3) = \infty$ (assuming that Turán's conjecture is true). However, determining $\mathrm{ex}(n, K_4^3)$ still seems far beyond reach. So we pose the following problem.

Problem 6.2. *Determine $ex(n,\mathcal{F})$ for some family $\mathcal{F}$ with $\xi(\mathcal{F}) = \infty$.*

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